Continual Learning With Memorized Dynamic Visual Description for Remote Sensing Scene Classification

Rui Peng, Wenzhi Zhao, Liqiang Zhang, Kun Jia, Qiao Wang, and Fengcheng Ji

***Abstract*—The continual updating of data renders the design of model architectures time-consuming and energy-intensive when tackling diverse tasks, and it quite commonly confronts weak generalization performance, even leveraging fine-tuning or transfer learning. Furthermore, the model tends to be well adapted for the current task while suffering performance degradation on previous tasks, namely, the catastrophic forgetting problem. Hence, there is an urgent demand to maintain models to learn new knowledge and alleviate knowledge forgetting, specifically balancing the plasticity and the stability of the model during the continual learning (CL) process. To address the aforementioned challenges, we propose a dynamic visual description and playback mechanism that leverages deep feature clustering and memory across diverse tasks. The former extracts stable, invariant features by clustering typical visual characteristics of scenes, accounting for spatial patterns, seasonal variations, and lighting conditions, thereby enhancing the model's comprehension of continuously evolving remote sensing scenes. The latter selectively memorizes features from specific samples to preserve historical model knowledge. This approach significantly strengthens the model's capabilities, deepening its understanding of scene imagery while mitigating the forgetting of historical knowledge. We generate several sequence tasks on AID, RSI, NWPU, SEASONET, and different scaled merged datasets to simulate the CL process across different resolutions, seasonal variations, and illumination changes. Compared to the classical CL methods, we demonstrate the superior performance of our model in dealing with continually updated remote sensing data.**

***Index Terms*—Continual learning (CL), dynamic clustering, remote sensing, scene classification, visual description.**

I. Introduction

THE rapid development of satellite technology has brought massive amounts of high-quality data to the remote sensing field, which is widely available to assist in understanding and interpreting the complex Earth's surface. Especially, the internal sophisticated geometrical structure and spatial configuration have caused significant challenges to the intelligent understanding of remote sensing images. Among the abundant very-high-resolution (VHR) remote sensing images, scene classification, which annotates image patches with semantic labels (e.g., residential areas, grasslands, and rivers) in terms of the internal objects and their distribution pattern, has achieved a rapid understanding of remote sensing images. Recently, remote sensing scene classification has attracted much attention in the remote sensing community and has been widely employed in image retrieval, land-cover classification [1], environmental management [2], and disaster monitoring [3].

With sufficient updated data, remote sensing scene images contain sophisticated combinations of internal objects and lead to a more complicated process of understanding these images. Specifically, remote sensing images may contain the same type of objects with nuance differences; for example, buildings, roads, and cars that are present within commercial scenes will also be present in residential, parking, viaduct, and other scenes. Moreover, the ever-updating nature of satellite data acquisition introduces a dual challenge to scene classification. First, the data stream involves class incremental, where new semantic categories (e.g., emerging land-use types) appear sequentially. Second, variations in illumination, seasonality, and sensor specifications cause domain-incremental problems, where distribution shifts [out-of-distribution (OOD)] occur even within the same scene category.

On the one hand, to recognize scene type with complex image objects, the powerful feature extraction capability of deep learning models ensures such tasks with state-of-the-art performances [4]. For instance, recent advancements have expanded these capabilities to complex interpretation tasks, establishing benchmarks for hyperspectral instance segmentation via spectral–spatial feature fusion [5] and introducing end-to-end frameworks for efficient raster-to-vector conversion [5]. To exploit intraclass variation and interclass similarity, some methods have employed deep convolutional neural networks (CNNs) to acquire representative features of VHR images [6]. Due to their multilayer structure, most CNN-based models understand scene images from the perspective of hierarchical feature extraction [7]. However, it remains difficult to obtain the global relationship between different objects. Consequently, for scene classification tasks, there is a significant demand to enhance the association between image objects across different categories to improve understanding during the continual learning (CL) process. Unlike CNNs, the Transformer is designed to capture the order and relationships

Received 21 September 2025; revised 19 January 2026; accepted 10 February 2026. Date of publication 17 February 2026; date of current version 12 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62471047 and Grant 62201063, in part by Beijing Natural Science Foundation under Grant L241048, and in part by the National Natural Science Foundation of China Major Program under Grant 42192580 and Grant 42192584. *(Corresponding author: Wenzhi Zhao.)*

Rui Peng, Wenzhi Zhao, Kun Jia, Qiao Wang, and Fengcheng Ji are with the Advanced Interdisciplinary Institute of Satellite Applications, Beijing Normal University, Beijing 100875, China (e-mail: wenzhi.zhao@bnu.edu.cn).

Liqiang Zhang is with the Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China.

Digital Object Identifier 10.1109/TGRS.2026.3665622

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

of elements in sequential data. Inspired by this structure, the Vision Transformer (ViT) captures global contextual information through image patching and the self-attention (SA) mechanism [8]. By dividing images into patches, the model establishes global relationships among internal unique elements, assisting in scene understanding. Nevertheless, standard ViT models typically rely on a static latent space. While effective for fixed datasets, they struggle to adapt to the aforementioned dual challenges, failing to accommodate the continuous influx of new classes without catastrophic forgetting or to generalize across domain shifts in dynamic environments.

On the other hand, to tackle distribution shifts, some researchers have leveraged transfer learning to engage the model in learning latent knowledge across different domains [9]. Pretrained models on natural images contribute notably when deep abstract features are adapted to remote sensing tasks, such as transferring pretrained ResNet or VGG from ImageNet (source domain) to remote sensing scene classification (target domain) [10]. Admittedly, the current CNN models trained on various natural image datasets have been effectively applied to image classification tasks in other fields. However, the significant variation between natural and remote sensing images causes domain shifting, meaning that the model cannot be immediately suitable for downstream remote sensing tasks. Furthermore, while some studies have attempted direct pretraining with numerous remote sensing images to acquire better feature representations [11], these approaches fundamentally remain static solutions. They focus on initial generalization but do not address the stability–plasticity dilemma required for continuous monitoring: they cannot update their knowledge base to incorporate newly emerging categories or adapt to seasonal variances without expensive retraining.

Crucially, when standard models adapt to novel data streams, the updating process often causes the erasure of previously learned information, a phenomenon widely known as catastrophic forgetting. Generally, deep neural networks struggle to maintain performance on historical tasks while sequentially adapting to new ones. To address this, CL, also known as lifelong learning, aims to achieve a delicate balance between plasticity—the ability to acquire and refine new knowledge from incoming data [12]—and stability—the capacity to preserve memories of historical knowledge [13], [14]. This balance is particularly vital for continuously updated remote sensing tasks, which demand the elimination of redundant information and the capture of highly representative descriptions. Although traditional approaches such as bag-of-visual-words (BoVWs) and CNN-based variants [14] have attempted to construct image representations using discrete "visual words," they often lack the capability to extract robust, invariant features or establish global relationships when confronted with nonstationary remote sensing data streams. Consequently, the key to successfully handling variably distributed data in a CL process lies in generating dynamically evolving visual descriptions that can adaptively represent complex scene images without succumbing to forgetting.

To address the dual challenges inherent in real-world remote sensing data streams—specifically the continuous emergence of new categories (class-incremental learning) and the distribution shifts caused by cross-temporal acquisitions (domain-incremental learning)—we present a novel framework named dynamic visual description with memory (DVDM). Unlike traditional methods that rely on static features, DVDM combines dynamic deep visual descriptions with a gradient-constrained memory mechanism to ensure robust knowledge retention. The model is fundamentally designed to bridge the semantic gap in nonstationary environments. First, we utilize continuously updated K-Means clustering to generate a set of representative visual words. These visual words function as a universal vocabulary, constructing stable visual relationships that bridge the semantic gap across different tasks and sensing phases. By describing scene images via these evolving visual patterns, the model enhances its comprehension of both novel classes and known classes under varying environmental conditions. Subsequently, to explicitly tackle catastrophic forgetting, we design a feature memory mechanism coupled with gradient projection constraints. This ensures that the optimization direction for new tasks does not conflict with the preservation of historical knowledge, effectively navigating the stability–plasticity dilemma. In summary, the main contributions of this article are given as follows.

1) We introduce a dynamic visual description mechanism that constructs a universal vocabulary via continuous clustering to overcome the limited expressiveness of static descriptors and capture evolving scene patterns.
2) We establish a unified framework capable of addressing the dual challenges of class-incremental and cross-temporal distribution shifts in remote sensing data streams.
3) We propose a dynamic feature memory mechanism coupled with gradient projection constraints, effectively resolving the stability–plasticity dilemma without catastrophic forgetting.

The rest of the components of this article are given as follows. Section II describes the related progress of remote sensing scene classification and CL. Section III introduces the proposed method in detail. Section IV provides the experimental data used in this article and the experimental setup. Section V shows the experiment results, and Section VI discusses the influence of module components and the effectiveness of the model. Section VII provides the main conclusion of this article.

II. Related Work

This section briefly summarizes the various types of remote sensing scene classification methods, as well as the groups of CL methods in recent years, which mainly consist of parameter regularization, parameter isolation strategy, and experience replay.

A. Remote Sensing Scene Classification

Remote sensing scene classification methods generally fall into three categories based on feature extraction strategies: low- [15], mid- [16], and high-methods [17]. Early

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

approaches relied on handcrafted low-level features, such as gray-level co-occurrence matrices (GLCMs) and color histograms. However, these methods depend heavily on manual feature engineering and often fail to capture high-level semantic information. While mid-level approaches such as BoVW attempted to bridge this gap, they still lack the capacity to fully describe complex spatial patterns. Consequently, the CNNs have recently become the dominant paradigm, offering superior capability in automatically learning hierarchical feature representations from raw imagery.

Within the CNN-based paradigm, existing methodologies typically adopt three primary strategies [4]: 1) as the feature extractor [18]; 2) fine-tuning [19]; and 3) full training [20]. Since traditional methods rely more on handcrafted features, some researchers, such as Cheng et al. [21], obtain visual words of scene images directly through the CNN model, and Zheng et al. [22] effectively implemented the scene classification task via multiscale pooling features of the last CNN layer and the Fisher vector (FV) method. Muhammad et al. [23] use a pretrained VGG model to extract mid-level and deep features, and incorporate SVM to implement the scene classification. Limited by the insufficient annotated samples, several researchers choose to handle remote sensing problems by utilizing a pretrained CNN model and fine-tuning the parameters. For example, Zhang et al. [24] designed the CNN-CapsNet that utilized a pretrained CNN to extract the representative features of the scene images and recognize images through the final CapsNet. As for the full training strategy, considering the disparity between natural and remote sensing images, some efforts involve designing and training new model structures from scratch.

However, these methodologies are predominantly predicated on the assumption of a closed-set scenario with offline batch training. Consequently, they struggle to handle the dynamic nature of real-world observation, specifically the nonstationary distributions characteristic of continuous remote sensing data streams. This rigidity leads to a fundamental stability–plasticity imbalance, where models fail to acquire new patterns without overriding previously learned knowledge, thereby restricting their deployment in long-term monitoring missions.

B. CL Methods

In contrast to biological systems that seamlessly accumulate knowledge, neural networks trained on nonstationary data streams are prone to catastrophic forgetting—a phenomenon where performance on historical tasks deteriorates as the model adapts to new data. To navigate the stability–plasticity dilemma, current CL methodologies are categorized into three paradigms [25]: 1) parameter regularization; 2) parameter isolation; and 3) experience replay.

Parameter regularization mitigates forgetting by constraining the update of crucial weights. Prominent methods such as elastic weight consolidation (EWC) [26] and memory-aware synapses (MASs) [27] introduce penalty terms to the loss function, estimating parameter importance via the Fisher information matrix and output sensitivity, respectively. To reduce the computational cost associated with the Fisher matrix, synaptic intelligence (SI) [28] employs an online surrogate loss to track importance accumulation. Beyond penalty terms, learning without forgetting (LwF) [29] leverages knowledge distillation to preserve learned representations, while AGS-CL [30] reinitializes less critical parameters to ensure sufficient model capacity for future tasks.

The parameter isolation strategies mitigate interference by assigning task-specific parameters, either by expanding the architecture or masking specific weights [31]. For instance, PackNet [32] iteratively prunes and freezes weights for previous tasks, freeing up the remaining capacity for subsequent learning. Similarly, Ostapenko et al. [33] employ a modular approach with local module composition to adapt to new tasks. Taking a bio-inspired approach, Wang et al. [34] coordinate a multilearner architecture by simulating the Drosophila olfactory system.

Experience replay methods maintain past knowledge by rehearsing either cached raw samples or generated pseudosamples. Rehearsal-based approaches such as iCaRL [35] and gradient episodic memory (GEM) [36] explicitly store a subset of historical data to constrain updates in future stages. To address storage limitations and privacy concerns, generative replay utilizes trained models to synthesize historical patterns. For example, Wu et al. [37] integrated a conditional GAN to reproduce seen samples, while Bang et al. [38] proposed rainbow memory (RM) to enhance sample diversity via augmentation. Furthermore, FeTrIL [39] combines a frozen feature extractor with a pseudofeature generator, balancing the learning of new classes with the recovery of past feature representations.

Specifically, within the remote sensing domain, recent studies have proposed several CL methodologies to navigate the inherent difficulties of scene understanding. Ammour et al. [40] addressed the dependence on past data by proposing a dual-branch architecture that couples a classifier with a data autogeneration subnetwork, enabling the model to synthesize historical samples from latent structures. Focusing on feature robustness, Peng et al. [41] introduced the continual contrastive learning network (CCLNet), which synergizes spatial and category-level knowledge distillation to mitigate forgetting, while employing contrastive learning to refine feature discriminability. To tackle label scarcity, Anwar Ma'Sum et al. [42] developed UNISA, an unsupervised flat-wide learning framework that extends few-shot CL to scenarios without annotated samples. More recently, in pursuit of parameter efficiency, Zhao et al. [43] explored prompt learning for scene classification, demonstrating that optimizing lightweight, learnable prompts can effectively facilitate continuous knowledge acquisition.

While contemporary CL algorithms effectively synergize multiple mechanisms to mitigate catastrophic forgetting, their performance remains bounded by the limited expressiveness of conventional deep feature descriptors. This inadequacy hinders the effective adaptation of remote sensing scene tasks over time. Consequently, the prerequisite for sustained comprehension of scene data streams lies in developing mechanisms capable of generating highly discriminative visual representations that accurately capture evolving patterns.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

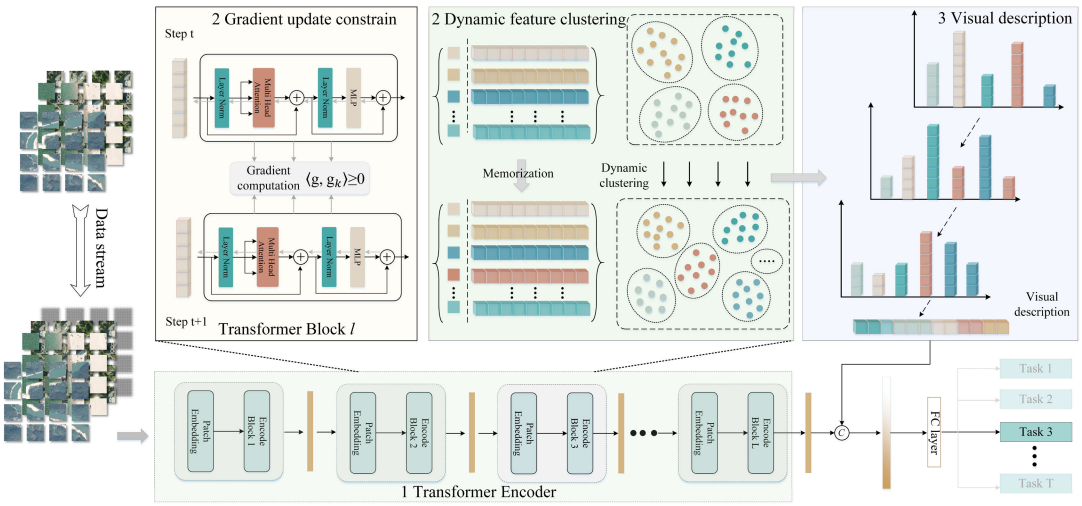


Fig. 1. Framework of our proposed method.

III. Methodology

In this section, we first formulate the problem of continual scene classification and define the associated notations. Subsequently, we detail the proposed framework, which consists of three integral components: 1) we utilize a ViT as the backbone to extract high-level token features from scene images; 2) we design a memory dictionary to retain representative historical features for dynamic clustering and employ gradient constraints to regulate the training of subsequent tasks; and 3) we leverage the cluster centers derived from the memory dictionary to generate dynamic visual descriptions, which are then concatenated with the backbone features for robust scene recognition. The overall framework is shown in Fig. 1.

A. Feature Generation

We assume in the CL procedure that the scene classification tasks are defined as $\{D_1, D_2, D_3, \ldots, D_t, \ldots\}$ $t \in T$ with the data consistently updated. Specifically, for a scene classification task at stage t, the samples can be denoted as $\{\ldots, (x_{i-1}^t, y_{i-1}^t), (x_i^t, y_i^t), (x_{i+1}^t, y_{i+1}^t), \ldots\} \in D_t$, where x_i^t denotes the i th scene image in the task at stage t and y_i^t is the corresponding semantic label. The image size could represent as $H \times W \times C$, which denotes the height, width, and channel, respectively. In general, the currently common visual large generative model adopts ViT as the backbone, while the image patching and SA mechanisms are applied to further improve the model performance. Therefore, here, we employ the ViT model as the backbone of the proposed framework for continual model training. For the remote sensing scene image x_i^t, it will be processed into fixed-size image patches, which is cropped into a sequence $\{(P_1^t, P_2^t, \ldots, P_n^t) | P \in \mathbb{R}^{p^2 \cdot c}\}$ specifically as t-stage model input, where P denotes a remote sensing scene image patch of size p × p and band c, and n is the length of the patches sequence and generally computed from $n = ((H \times W)/p^2)$. Subsequently, the above patch sequence is flattened and linearly projected to the representative vector features with d-dimension using the learnable embedding matrix $E \in \mathbb{R}^{p^2 \cdot c \cdot d}$, concatenating a learnable category token y_c^t with these embedding features for model training. To keep the positional information of the patch sequence, the vector features additionally encoded the position embedding $E_p \in \mathbb{R}^{(n+1) \cdot d}$ as the model input simultaneously. The input embedding could be expressed as follows:

$$z_{input}^t = \left[y_c^t; P_1^t E; P_2^t E; \ldots; P_n^t E\right] + E_p. \tag{1}$$

The acquired embedding features z_{input}^t are then fed into the Transformer encoder at t-stage, which consists of L stacked blocks. Each block consists of the multihead SA (MSA) layer and the multilayer perceptron (MLP) layer: the former is used to interact with global tokens in the encoder, the latter is used to provide nonlinear and localized information, and both of them collectively enhance the performance of the model in handling the scene classification task. Thus, the feature extraction of the input scene image could be formulated as follows:

$$z_l^{t\prime} = MSA\left(LN\left(z_{l-1}^t\right)\right) + z_{l-1}^t, \quad l \in (1, 2, \ldots, L) \tag{2}$$

$$z_l^t = MLP\left(LN\left(z_l^{t\prime}\right)\right) + z_l^{t\prime}, \quad l \in (1, 2, \ldots, L). \tag{3}$$

The MSA is applied to evaluate the importance of the embedding features with other embedding features in the Transformer encoder; especially when the model is deeper, the SA mechanism allows the model to more easily focus on the embedding features that are farther away inside the embedding sequence of the scene image. In general, the SA mechanism produces query, key, and value, which are indicated by Q, K, and V respectively, by projecting the input sequence and multiplying them with three initialized trainable matrices W^Q, W^K, and W^V. The query vector and the key vector measure the similarity of each element in the input sequence by dot product and then divide it by the scaled key value D_K to get the weighted information that incorporates the other elements, enabling the interaction of contextual information. The detailed process can be expressed as follows:

$$[Q, K, V] = zU_{QKV} \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

$$SA(z) = softmax\left(\frac{QK^T}{\sqrt{D_K}}\right) \cdot V. \tag{5}$$

To further enhance the mode's ability, the *MSA* block in VIT acquires multiple sets of corresponding weight matrices, and it allows each head to focus on different aspects to capture richer information. The formula could be expressed as follows:

$$\begin{aligned} MSA(z) &= Concat(h_1, h_2, \ldots, h_h)\, W \\ h_i &= SA\left(zW_i^Q, zW_i^k, zW_i^V\right) \end{aligned} \tag{6}$$

where h denotes the number of model-defined heads, and W_i^Q, W_i^k, and W_i^V are the corresponding weight matrices in each head.

B. Memory Mechanism and Feature Description

To preserve the model's memory of the previous tasks during CL, we kept some scene images from the current task, as well as the deep-patched scene features. The ViT encoder provides excellent representative deep features for the current task. Thus, we defined two memory dictionaries for storing portions of scene images and feature representations. The memory operation of the t-stage M^t can be represented as follows:

$$\begin{aligned} M^t &= \left\{\left(x_1^t, y_1^t\right), \ldots, \left(x_m^t, y_m^t\right) | Dict^t\right\}, \quad m = \frac{M}{T} \\ Dict^t &= \left(z_1^t, z_2^t, z_3^t, \ldots, z_m^t\right) \end{aligned} \tag{7}$$

where x_m^t represents the mth saved scene image, $Dict^t$ saved the visual features of scene images, and z_m^t represents the specific encoded patch features, according to the size of the patch window $z_m^t \in \mathbb{R}^{n \times d}$, where M denotes the capacity of the memory dictionary and T is the length of the task sequence, whereas, in the open scenario, it would be direct to set m as the number of samples stored per stage of the task for uncertain task length.

In the CL process, the same object often exists across different scene images. Therefore, to further enhance the model's ability to understand scene images, we additionally leverage the internal feature representation of the scene, i.e., M^t, collected in the task stream to describe the scene images. Specifically, all feature vectors collected in the t-stage task are clustered using K-Means. To ensure that the generated centers constitute stable and representative visual descriptions, the clustering process aims to minimize the within-cluster sum of squares. The objective function J is formally defined as

$$J = \sum_{j=1}^{K} \sum_{f_i \in S_j} \left\| f_i - c_j \right\|^2 \tag{8}$$

where S_i represents the set of feature vectors assigned to the jth cluster. The cluster centers are iteratively updated, specifically by computing the mean of the assigned vectors

$$c_j = \frac{1}{|S_j|} \sum_{f_i \in S_i} f_i. \tag{9}$$

Through this optimization, we obtain a set of robust cluster centers $C = \{c_1, c_2, c_3, \ldots, c_K\}$, as shown in Fig. 2. These centers serve as the base visual words, which are applied to describe the scene images. Thus, the cluster centers at t-stage are considered as base words, which are applied to describe the scene images

$$v_k = \sum_{i=1}^{n} \mathbb{I}\left(\arg\min_j \left\| f_i - c_j \right\|^2 = k\right) \tag{10}$$

where v_k denotes the frequency of the kth visual word, which counts the deep patch features f_i that are closest to the cluster center c_k. Here, n is the total number of extracted deep features, and c_j represents the jth cluster center. Thus, the description for the scene image can be expressed as

$$z_f^t = \left[v_1^t, v_2^t, \ldots, v_K^t\right]. \tag{11}$$

Subsequently, the acquired feature description vectors are fused with the previously encoded features z_l^t in (3), the encoded features improved the context information of scene acquisition, and the description features enhanced the understanding of the interior of the scene during the CL process of the model. Therefore, the model performed well by integrating these two essential pieces of information. Finally, the encoded features and description features are input into the final classification layer for scene classification, and this fusion process can be expressed as

$$z_{in} = Concat\left(z_l^t, z_f^t\right). \tag{12}$$

C. Gradient Updating With Memory Dictionary

To keep the current t-stage model update affecting the previous stage's model parameters as minimally as possible, it is required to ensure that the loss value of the previous task does not increase during the model updating process. The loss function at t stage is defined as

$$L\left(\theta, M^t\right) = \frac{1}{|M^t|} \sum_{(x_i, t, y_i) \in M^t} L\left(f\left(x_i^t, z_i^t, t\right), y_i\right) \tag{13}$$

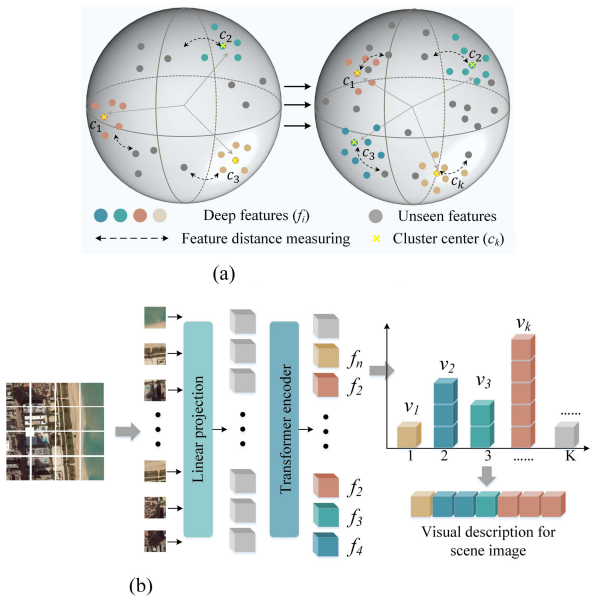


Fig. 2. (a) Dynamic clustering and (b) visual description generation process during CL.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Continual Learning With Dynamic Visual Description and Gradient Constraint

Require: Sequential tasks T = $\{D_1, D_2, \ldots, D_T\}$; ViT Backbone parameters θ; Memory Dictionary M (initially $\emptyset$); Cluster number K; Learning rate η.

Ensure: Final updated model parameters θ_T.

for task $t = 1$ to T **do**
  # Update clustering centers
  if $\mathcal{M} \neq \emptyset$ **then**
    Collect features from $\mathcal{M}$ and current dataset D_t
    Perform K-Means to obtain cluster centers C = $\{c_1, c_2, \ldots, c_K\}$
  else
    Initialize cluster centers C using D_1
  end if
  for each batch (x_{batch}, y_{batch}) in D_t **do**
    # Feature extraction and description generation
    Visual word frequency $v_k = \mathbb{I}\sum_{i=1}^{n}\left(\arg\left\|f_i - c_j\right\|^2 = k\right)$
    Construct description vector $z_f^t = \left[v_1^t, v_2^t, \ldots, v_K^t\right]$
    Concatenate features: $z_{in} = Concat\ (z_l^t, z_f^t)$
    # Loss calculations
    Compute predictions $\hat{y} = Classifier\ (z_{in})$
    Calculate loss $L_{task} = L_{CE}\ (\hat{y}, y_{batch})$
    Compute gradient $g = \nabla\theta L_{task}$
    # Gradient updates with constraints
    if $t > 1$ **then**
Compute reference gradients g_{ref} using samples in $\mathcal{M}$
      if $\langle g, g_{ref} < 0\rangle$ **then**
        Project gradient g to $\hat{g}$ subject to $\langle \hat{g}, g_{ref}\rangle \geq 0$
        Update parameters $\theta \leftarrow \theta - \eta \cdot \hat{g}$
      else
        Update parameters $\theta \leftarrow \theta - \eta \cdot g$
      end if
    else
      Update parameters $\theta \leftarrow \theta - \eta \cdot g$
    end if
  end for
  Update $\mathcal{M}$ with representative samples from D_t
end for

where θ is the parameter of the model and M^t is the memory dictionary, which saves part of the scene images and corresponding visual features during the CL process

$$L\left(f_\theta^t\left(x_i^t, z_i^t, t\right), M^k\right) \leq L\left(f_\theta^{t-1}, M^k\right) \quad for\ all\ k < t. \tag{14}$$

For maintaining the memory of historical knowledge in a continual scene classification task, here, we adopt gradient constraints while memorizing historical samples and features to control the forgetting of historical knowledge in the current phase of the model

$$\langle g, g_k\rangle := \left\langle \frac{\partial L\left(f\left(x^t, t\right), y\right)}{\partial\theta}, \frac{\partial L\left(f, y\right)}{\partial\theta}\right\rangle \geq 0 \quad for\ all\ k < t \tag{15}$$

where $\langle\cdot\rangle$ indicates the angle between two loss gradient vectors. Once the current gradient update direction of the model is acute with the previous task gradient update direction, then the model is considered to have effectively preserved the knowledge acquired by the previous model. When the model updating process satisfies the condition of the gradient change angle of all tasks, the losses of these tasks usually tend not to change, and only when the gradient direction of one or more tasks violates (15) during the updating process, the loss of previous tasks will increase, and then, the gradient g can be projected to the closest gradient $\hat{g}$; hence, the goal of the model's gradient updating changes to

$$\begin{aligned} &minimize_{\hat{g}}\ \frac{1}{2}\|g - \hat{g}\|_2^2 \\ &subject\ \langle \hat{g}, g_k\rangle \geq 0 \quad for\ all\ k < t \end{aligned} \tag{16}$$

where $\hat{g}$ denotes the g closest gradient in squared L2 norm, and when the task model is updated, the constraints can be changed in such a way as to place $\hat{g}$ in an acute angle with the previous task vectors.

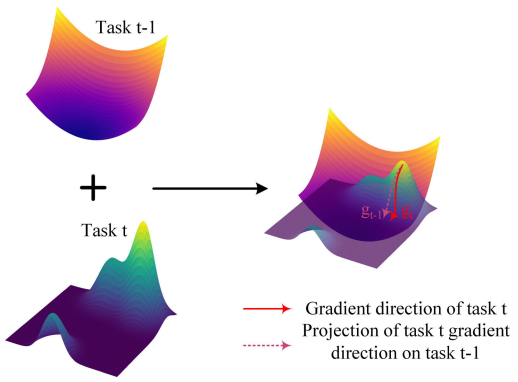


Fig. 3. Gradient update direction of the model.

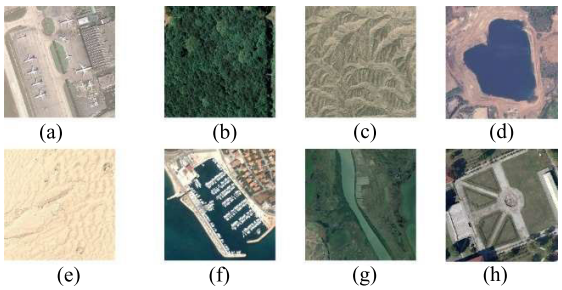


Fig. 4. Scene samples of the AID dataset. (a) Airport. (b) Forest. (c) Mountain. (d) Pond. (e) Desert. (f) Port. (g) River. (h) Square.

As shown in Fig. 3, it is possible to make the projection of the gradient update direction of the model at stage t match the direction of the gradient descent at stage t-1 through the constraints on the update of the model gradient, which effectively preserves the memory of the historical knowledge. The complete procedure of the proposed DVDM is summarized in Algorithm 1.

IV. Dataset and Experiment Setting

A. Datasets

1) AID: The AID dataset [44] is a high-resolution optical aerial image provided by Wuhan University and Huazhong

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

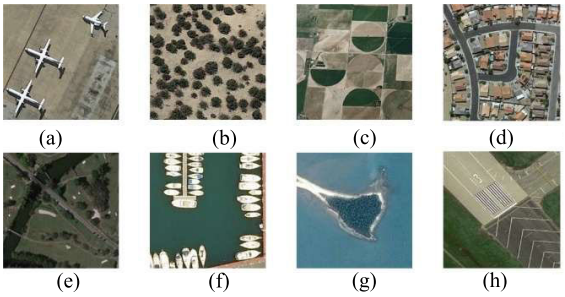


Fig. 5. Scene samples of the NWPU dataset. (a) Airplane. (b) Chaparral. (c) Circular farmland. (d) Dense residential. (e) Golf course. (f) Harbor. (g) Island. (h) Runway.

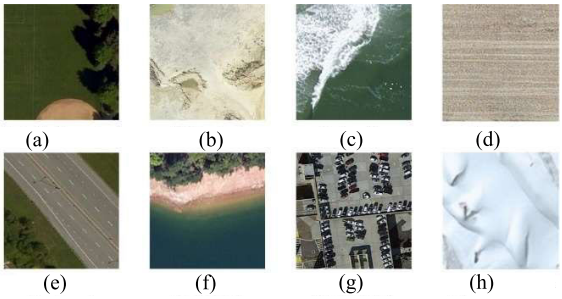


Fig. 6. Scene samples of the RSI-CB dataset. (a) Artificial grassland. (b) Bare land. (c) Coastline. (d) Dry farm. (e) Highway. (f) Lakeshore. (g) Parking lot. (h) Snow mountain.

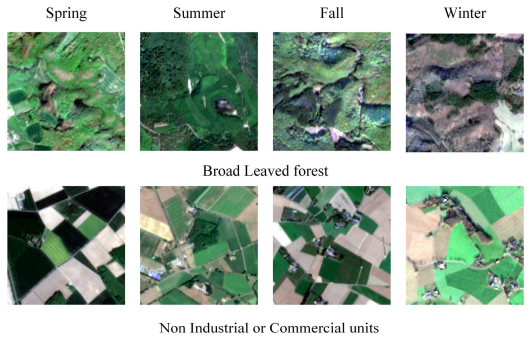


Fig. 7. Scene samples of the SEASONET dataset in different seasons.

University of Science and Technology (HUST), which provides a total of 10 000 images with labels covering 30 different scene categories, including Airport, Bare Land, Commercial, Meadow, Park, and so on; each category has 200–420 images with spatial resolution from 0.5 to 8 m, and the image size is 600×600 pixels. Partial samples from this benchmark dataset are exhibited in Fig. 4.

2) NWPU-RESISC45: The NWPU-RESISC45 [45] dataset is a large-scale benchmark scene dataset provided by Northwestern Polytechnical University, the total number of images in this dataset reaches 31 500, containing 45 different scene classes, such as Airplane, Beach, Church, and Forest, each class has 700 images with spatial resolution from 0.2 to 30 m, and the image size is 256×256 pixels. Similarly, parts of the samples are illustrated in Fig. 5.

3) RSI-CB: The RSI-CB [46] dataset is a benchmark dataset provided by Central South University, and according to the requirement of the model on the input data size, it is designed with images of 256×256 pixel size and 128×128 pixel size; here, we chose to utilize the former for the following experiments, which included a total of 24 000 images with spatial resolution from 0.3 to 3 m in 35 different categories, such as Airplane, Bridge, Container, Hirst, and Mountain. Some of the scene samples are exhibited in Fig. 6.

4) SEASONET: The SEASONET [47] dataset is contributed by the Pattern Recognition for Embedded Systems Group and is primarily designed for image segmentation and retrieval. It contains 1 759 830 Sentinel-2 tiles with 33 land-cover labels. Each image contains 12 bands with 128×128 pixel resolution. Since this dataset is designed for segmentation tasks, each pixel has a corresponding label. To adapt it for scene classification, we filtered these data through the dominant land-cover class, where the specific category occupies more than 60% of the image. Finally, we selected ten different scene categories with 300 RGB image tiles, and to capture variations in illumination and seasonal conditions during the CL task, we collected each category to cover four seasons: spring, summer, fall, and winter, so each scene category finally contains 1200 images, and the whole dataset contains 12 000 images in total. Some of the scene samples with different season conditions are exhibited in Fig. 7, and the detailed data split for cross-temporal learning is displayed in Table I.

B. Experimental Setting

To simulate as much as possible the continual updating data flow of the real world, we divided the dataset internally into ten phases of different tasks, and we set three to five different categories in each phase, considering that the above three datasets contain different data categories when dividing the CL tasks. For example, for the AID dataset, the first stage contains Airport, Bare land, Baseball field, and three new categories of Beach, Bridge, and Center are added to the second stage task and so on. Subsequently, to simulate more complex real-world scenarios and evaluate long-term stability, we fused the three source datasets to construct two sophisticated composite benchmarks, referred to as Merge-80 and Merge-90. Specifically, by aggregating distinct scene categories from the aforementioned datasets, we formed two consolidated sets containing 80 and 90 classes, respectively. These large-scale benchmarks are then organized into longer task streams (up to 15 sequential tasks) to rigorously test the model's ability to deal with complex real-world scenes without catastrophic forgetting.

As mentioned previously, as shown in Table II, samples, about only 2% for parameter updating to better imitate the shortage of the number of samples under realistic conditions, and another 20% of the samples are selected for validation. In this article, ViT is employed as the backbone network of our method. The input image size is adapted to 224×224 pixel size by random scaling and cropping. Since a sequence of vision patches input is requested for the ViT model, we chose to crop the scene image using a patch size of 16×16 pixels and then compress it into a sequence as model input. In addition, we set the memorized samples and corresponding cut features in each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

TABLE I

SELECTED SCENE SAMPLES IN DIFFERENT SEASONS, AND THE TRAINING AND EVALUATION RATIO FOR CL IN A LARGER TIME SPAN. THE TR_L REPRESENTS TRAINING SAMPLES FOR LARGE-SCALE CL, AND THE TR_S AND TE REPRESENT THE TRAINING AND TEST SAMPLES OF FOUR SEASONS FOR CROSS-SEASON TASKS

Category	Seasons					TR_L	TR_S	TE
	Spring	Summer	Fall	Winter	Total			
Mixed_forest	300	300	300	300	1200	24	240	240
Non_industrial_or_commercial_units	300	300	300	300	1200	24	240	240
Intertidal_flats	300	300	300	300	1200	24	240	240
Broad_leaved_forest	300	300	300	300	1200	24	240	240
Coniferous_forest	300	300	300	300	1200	24	240	240
Pastures	300	300	300	300	1200	24	240	240
Coastal_lagoons	300	300	300	300	1200	24	240	240
Vineyards	300	300	300	300	1200	24	240	240
Sea_and_ocean	300	300	300	300	1200	24	240	240
Discontinuous_urban_fabric	300	300	300	300	1200	24	240	240

TABLE II

DATASET DESCRIPTION AND TASK SPLIT

Dataset	Image size	Tasks split	Tr / Class	Te / Class
AID	600*600	[3,3,3,3,3,3,3,3,3,3]	13-25	44-84
RSI	256*256	[3,3,3,3,3,4,4,4,4,4]	7-53	39-266
NWPU	256*256	[4,4,4,4,4,5,5,5,5,5]	14	140
Merge-80	/	[8,8,8,8,8,8,8,8,8,8]	5-14	52-144
Merge-90	/	[6,6,6,6,6,6,6,6,6,6,6,6,6,6,6]	29-38	292-384

phase to be 50 in the CL process and train 50 epochs in each task. As for the optimization algorithm, we chose Adam and set the initial learning rate to be $1e^{-5}$. To acquire robust results, we repeated all the experiments three times, and the time data were randomly selected. All experiments were performed on the Ubuntu 20.04.1 operating system; all code was written in Python version 3.8, PyTorch version 1.8.1, and the NVIDIA A800 GPU with 80-GB memory.

C. Evaluate Method

To evaluate the performance of our proposed method with other benchmark methods, we decided to utilize the conventional overall accuracy (OA). Besides, to estimate the knowledge transfer process of the model during the CL task, we elected to utilize backward transfer (BWT) and task average accuracy (TAA) to evaluate the model's memory for historical knowledge and its influences on unseen scenes. Specifically, BWT evaluates the influence of the current t-stage model trained on the previous task $k, k < t$, while TAA assesses the influence of the current t-stage model on current tasks to evaluate the stability of the model. As such, we evaluate all T stages of the task after each task is completed and obtain the accuracy matrix $A \in \mathbb{R}^{T,T}$, where $A_{i,j}$ represents the evaluated accuracy of the model on the jth stage after the ith stage training. In addition, setting the evaluated accuracy of the random initialization model at each stage to $\bar{b}$, then the above two evaluations can be expressed as

$$BWT = \frac{1}{T-1}\sum_{i=1}^{T-1} A_{T,i} - A_{i,i} \tag{17}$$

$$TAA = \frac{1}{N_T}\sum_{i=1}^{N_T} A_{i,T}. \tag{18}$$

Among them, a larger BWT indicates that the model remembers the knowledge of the previous task better; in other words, it effectively mitigates the catastrophic forgetting problem. Because of the nature of the BWT metric, we do not calculate the BWT for the first task.

V. RESULTS

A. Comparison With Different Methods

To evaluate the effectiveness of our proposed method, we compare it against a comprehensive set of representative CL algorithms. These baselines cover mainstream strategies including parameter regularization, parameter isolation, and experience replay, as well as recent state-of-the-art architectures. Specifically, the compared methods include ViT-CL [48], EWC, XdG [49], LwF, SI, A-GEM [50], DERPP [51], Co2L [52], and SimpleCIL [53].

For a fair comparison, the parameter settings for all baseline methods were standardized where possible. The initial learning rate was set to 0.001, and the batch size was set to 32. All models were trained for 50 epochs per task, with the exception of ViT-CL and Co2L, which required an additional 50 epochs for the pretraining phase. All other hyperparameters followed the default configurations reported in their respective original papers. Data augmentation strategies were kept consistent with our proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION EXPERIMENTS OF THE FEATURE DESCRIPTION LENGTH

l	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Task9	Task10
Base	36.38±0.99	41.58±0.86	59.50±2.65	65.88±3.13	35.80±6.00	39.36±4.01	34.67±6.18	51.00±2.27	77.38±11.09	45.26±2.62
64	69.07±0.60	63.51±1.38	90.50±1.34	90.57±3.67	76.52±1.78	74.51±2.77	82.44±4.08	82.67±2.72	93.54±2.10	82.28±1.08
96	68.43±2.55	63.51±1.74	89.60±0.25	90.09±3.67	77.25±1.82	72.27±3.48	86.22±2.20	80.67±2.72	96.94±0.83	80.35±2.86
128	**70.19±3.21**	**64.21±1.14**	91.76±0.91	90.56±1.68	76.81±2.36	**74.93±4.61**	85.11±2.57	82.50±4.71	**97.62±0.64**	79.30±3.28
256	69.39±4.41	62.81±2.73	**92.29±1.66**	**91.35±1.82**	**82.61±1.88**	74.51±4.19	**86.67±0.54**	**83.50±3.34**	97.11±0.64	**84.39±2.12**

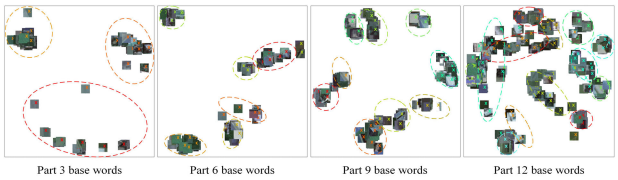


Fig. 8. Base words' visualization in feature space.

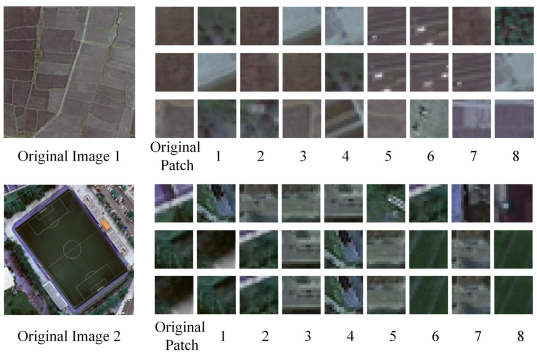


Fig. 9. Utilize the visual words to represent the scene image. (Left image) Original scene image. (Right) Patches represent the associated patches that use the 1–8 (here, we select 8 for demonstration) patches to describe the original patch. (b) Scene image patches with their visual description.

B. Ablation Study

1) Impact of Visual Description Length and Comparison Visualization Features: To verify the effectiveness of our proposed visual description for scene classification tasks, we performed a series of ablation experiments on the length of feature descriptions. We set a CL task on the AID dataset and split it into ten stages, with three different scene categories in each stage. The results of the ablation experiments are shown in the following figure and table.

The different feature lengths indicate the maximum length of the encoded feature descriptions of the scene images during the CL process, and we finally provide four different feature description lengths and a base model without any description for the ablation experiments. As shown in Table III, the longer length of the feature description brings better accuracy performance of the model. Generally, the 128- and 256-length feature descriptions acquired maximized accuracy performance in most steps of the task stream. The former length of feature descriptions performs better at later stages in the CL task stream, and the latter performs better in the upfront task. As shown in Fig. 8, where we visualized each patch and its corresponding deep features in the feature space, and where each color denotes that the basic visual word belongs to the same label, it can be found that more visual words

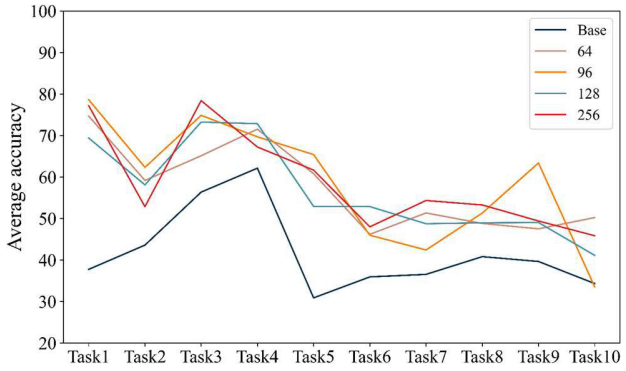


Fig. 10. Average accuracy of baseline and different feature description lengths.

provide a great help for the understanding of the scene image. In using visual word representations to represent scene images, as shown in Fig. 9, not only their own deep features are utilized but also patch features similar to them are utilized to provide more robust performance for CL.

Fig. 10 demonstrates the average accuracy of the model for all seen or unseen tasks at each stage. Comparing the different lengths of feature descriptions, most of them show a significant performance decrease of the model from task 2 to task 4, and the average accuracy exhibits weak stability in stages 6–10, whereas setting the feature description length to 128 achieves higher accuracy in most of the stages. It reveals that a feature description with a length of 256 can provide relatively acceptable accuracy performance for the models' sequential learning. Eventually, we decided to implement all our experiments with a 256-length feature description. The experiments proved that introducing additional visual descriptions can effectively improve the comprehension of scene images during CL.

In addition, we demonstrate the capability of recalling latent learned knowledge about scene images between our proposed model and other comparative methods in the streaming tasks. Specifically, we use t-SNE to transform the extracted high-level abstract features into two dimensions at different learning tasks, which is utilized to exhibit the discriminating capability of the model on learned data. As shown in Figs. 11–13, we display the visualization results from the model on the initial Task 1, as well as at the final Task 10 on the first step data, which is based on the three datasets, AID, RSI, and NWPU, respectively. In these figures, ViT and ViT-CL methods extracted highly discriminative features on the three datasets, but, according to the previous accuracy tables and the BWT and TAA metrics, these methods are failing to connect the features to previously acquired knowledge. As for our proposed method, it is still able to keep a better

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

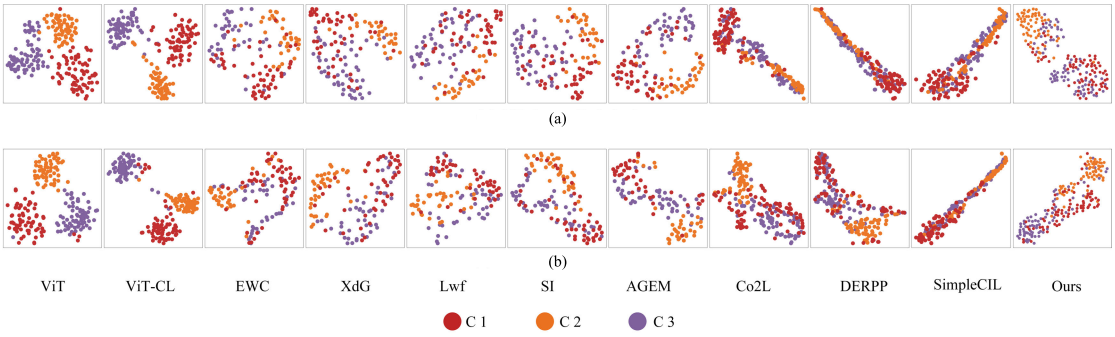


Fig. 11. t-SNE visualization on the AID dataset. (a) AID dataset Task 1-1. (b) AID dataset Task 10-1.

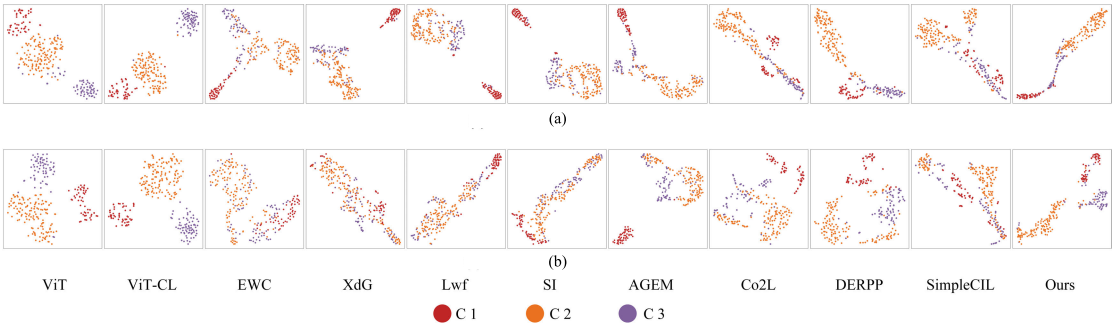


Fig. 12. t-SNE visualization on the RSI dataset. (a) RSI dataset Task 1-1. (b) RSI dataset Task 10-1.

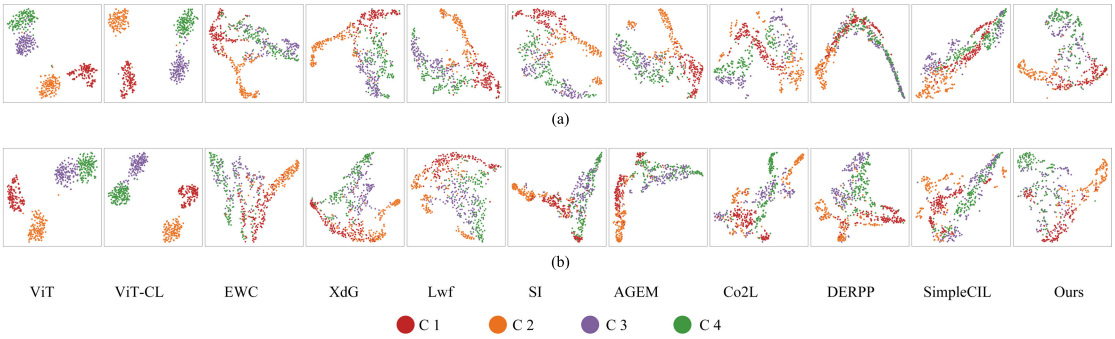


Fig. 13. t-SNE visualization on the NWPU dataset. (a) NWPU dataset Task 1-1. (b) NWPU dataset Task 10-1.

TABLE IV
COMPARISON RESULTS ONTHE AID DATASET

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Task9	Task10
ViT	90.71±1.38	77.89±0.74	96.59±2.5	95.28±1.02	93.91±1.55	95.8±0.34	90.67±1.89	96.0±2.16	98.81±0.48	94.39±1.51
ViT-cl	92.15±1.98	78.77±3.88	95.88±1.34	98.11±1.02	94.35±0.71	96.08±1.43	92.22±3.86	96.67±2.05	97.96±1.1	92.46±2.21
EWC	71.87±4.60	59.37±4.60	79.69±4.18	86.20±2.05	77.60±4.78	72.66±4.18	79.95±2.65	73.96±3.21	95.31±0.00	71.09±2.55
XdG	71.09±0.64	**64.85±3.37**	84.89±3.01	87.76±2.05	81.77±1.95	71.61±4.61	78.65±2.05	79.43±7.92	95.05±0.74	78.64±3.01
LwF	69.79±1.48	63.28±3.55	84.90±1.95	89.06±1.69	81.77±2.24	71.88±6.08	81.25±2.78	79.95±5.12	95.57±0.37	78.12±1.11
SI	70.57±0.98	63.54±3.51	84.11±1.33	88.54±0.37	83.07±2.05	69.53±5.85	82.55±0.37	79.95±4.83	93.49±2.24	78.64±1.95
AGEM	71.87±5.85	57.81±4.18	79.17±4.24	86.98±0.97	77.60±5.43	69.79±4.62	**87.76±2.58**	76.82±4.15	92.97±4.42	72.91±2.58
Co2L	72.92±1.59	42.98±0.89	63.62±4.51	47.64±0.0	65.8±2.17	40.2±11.72	39.33±6.06	47.5±16.28	74.83±3.07	55.79±5.23
DERPP	**79.01±4.88**	64.21±2.39	82.26±7.06	76.89±0.67	**89.13±5.58**	**96.92±2.28**	77.33±5.44	80.67±9.85	93.03±5.27	63.51±6.99
SimpleCIL	62.02±2.57	24.38±0.66	37.45±2.42	16.98±5.39	40.14±1.96	9.52±2.58	10.89±4.37	4.67±0.94	35.88±1.2	2.63±0.86
Ours	69.39±4.41	62.81±2.73	**92.29±1.66**	**91.35±1.8**2	82.61±1.88	74.51±4.19	86.67±0.54	**83.50±3.34**	**97.11±0.64**	**84.39±2.12**

discriminating ability for the past history data and even after several new steps of updating. Specifically, compared to the other methods, our approach performs relatively better in clustering the different categories of the tenth updated model for the first-stage data, and in particular, on the RSI dataset, where the feature distributions of the compared methods are closer together, the categories are in greater confusion. As for the NWPU dataset visualization, our method is superior

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

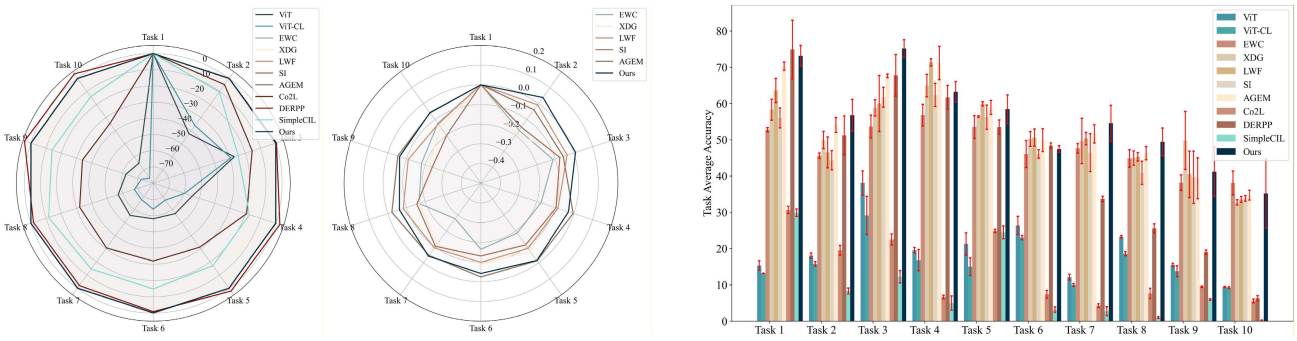


Fig. 14. BWT and TAA for different methods on the AID dataset. The red line represents the std value. (a-1) Ten-task BWT on the AID dataset. (a-2) Ten-task BWT on the AID dataset. (b) Task average accuracy comparison on the AID dataset.

in identifying the C1, C2, and C4 categories, whereas C3 is easily mixed up with other categories like all other methods.

C. Comparison Experiments

1) Comparison Experiment on the AID Dataset: Table IV shows our comparison of two regular scene classification methods and seven other typical CL methods on the AID dataset in a ten-stage CL task. Compared to the ViT and ViT-CL methods, it is clear to notice that numerous CL methods perform weakly on the current stage of the task, which is due to the fact that ViT- or ViT-based models are able to extract significantly discriminative features. Comparing with these CL methods, our method is able to learn new categories efficiently in long sequences of CL tasks, which, in turn, leads to the best classification results.

Among these ten different stages, our method achieves the best performance in the majority of the stage tasks. While the DERPP, XdG, and AGEM methods achieve better results only in stages 1, 2, 5, 6, and 7, our proposed method only falls short of them by 9.62%, 8.52%, 22.41%, and 1.09%, respectively. In facing the newest classification task of the newest category, in all the remaining stages, the accuracy of our proposed method compared to the other methods yielded higher results; specifically, in stage 3, we improved by 13.12% of the accuracy compared to the AGEM, and in Task 6, we improved 4.98% of the accuracy compared to the SI; especially, in Task 9, where all the methods reached more than 90% of the accuracy, the accuracy of our method is still improved by 4.14% compared to the AGEM method.

In order to demonstrate the stability of the model's learning ability at each stage of the CL process, we present the average task accuracy of the model at each stage, as shown in Fig. 14(b); our method achieves high average task accuracy in most stages; it is only slightly weaker than the comparative methods in fourth, sixth, and seventh tasks; and only the XDG and EWC methods perform slightly better in Task 9 and 10, respectively, In general, the average task accuracies in the figure indicate that our proposed model is able to maintain the stability of the model's learning ability in a series of long CL tasks, which makes our model perform more robustly in the CL.

As evident from the table, although the ViT and ViT-CL models achieved the highest accuracy in all tasks, in comparing the models in terms of their ability to memorize historical knowledge, these models obviously performed weaker than the CL methods. Specifically, to measure the influence of model updates on the historical tasks during CL for each model, we exhibit the variation of the BWT metrics of the models at each task. From Fig. 14(a-2), we observe that our method (black line) and AGEM method keep a higher BWT score, which indicates that our method and AGEM method memorize the essential knowledge stably that is important to historical data, thus effectively mitigating the catastrophic forgetting for sequence tasks, whereas the ViT, ViT-CL, and Co2L demonstrated significant forgetting on former tasks, which is shown in the green lines in Fig. 14(a-1).

In Fig. 15, we utilize the confusion matrix to illustrate the effect of the trained models of the different methods on the future tasks (top row), as well as the memory of the trained models on the historical tasks (bottom row), respectively. Other approaches tend to predict into the same category when confronted with data in future data streams, whereas our method is able to discriminate the data in future data streams to a certain extent. On facing historical data, compared with other methods, our method sustains the ability to memorize learned knowledge, which is more important for CL.

2) Comparison Experiment on the RSI Dataset: Over the former AID dataset, the RSI dataset contains a larger variety of scene categories and numbers of images, and there is a greater influence of the internal complexity of the scenes on the CL tasks, while Table V demonstrates our comparison of other methods on the RSI dataset in a ten-stage CL task. As can be seen from the table, our method is also able to achieve stable and best classification results in long sequential CL tasks except ViT and ViT-CL for the latest task. The ViT and ViT-CL keep a stable, high performance on current scene classification tasks and even obtain 100% at Task 10. Among these CL methods in ten different stages, our method achieves the best performance in the vast majority of stage tasks.

In the RSI dataset, Co2L and SI methods achieve better results only in stages 1, 2, and 5, where our proposed method only falls short by 5.75%, 4.65%, and 0.60%, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

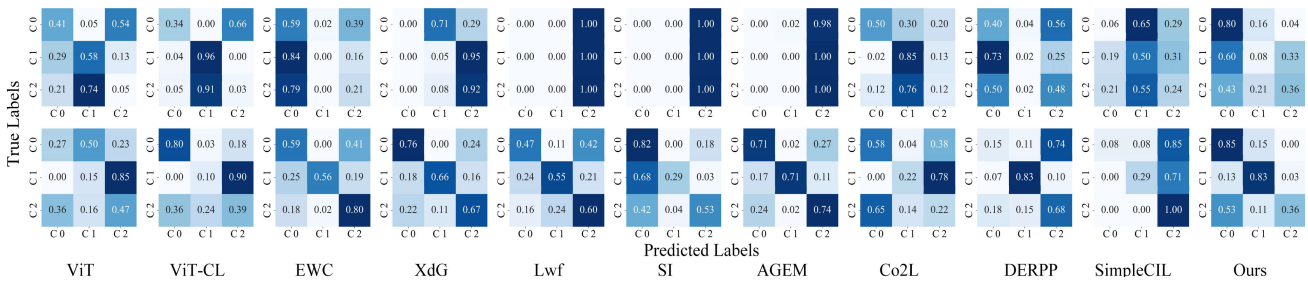


Fig. 15. Confusion matrix between our method and the comparison method on the AID dataset, where the first row represents the classification result of the first stage-trained model on the tenth data, which is utilized to indicate the discrimination ability of the model on the future data, and the second row represents the classification result of the Task 10 trained model on the data in first split dataset, which is utilized to estimate the memorization capacity of the model on the historical data.

TABLE V
COMPARISON RESULTS ONTHE RSI DATASET

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Task9	Task10
ViT	94.13±1.01	98.21±0.25	98.35±0.19	98.55±0.43	99.72±0.28	97.42±1.22	92.73±1.87	96.74±0.69	99.2±0.41	100.0±0.0
ViT-cl	94.38±2.01	99.02±0.63	98.67±0.4	98.3±0.6	99.83±0.13	98.49±0.63	90.37±2.48	96.81±0.26	99.32±0.37	100.0±0.0
EWC	83.53±1.36	93.30±0.22	86.82±0.38	94.73±1.05	98.02±0.24	93.84±1.33	80.47±5.28	85.14±1.32	94.48±2.31	94.81±3.02
XdG	85.57±0.91	93.83±1.36	90.74±1.66	94.73±1.48	97.80±0.16	95.20±0.27	85.10±3.92	88.40±1.97	97.64±0.38	95.34±0.83
LwF	86.21±1.56	94.37±0.44	89.49±1.63	94.39±1.27	97.91±0.41	94.77±0.37	85.29±2.48	89.44±0.86	96.12±0.74	94.13±0.48
SI	86.21±1.88	94.19±0.77	89.73±0.95	94.30±2.29	**98.40±0.28**	95.05±1.07	86.10±3.32	88.47±0.80	97.79±0.47	95.34±0.72
AGEM	87.99±1.10	93.57±1.31	84.94±3.74	93.20±1.07	89.31±4.89	90.11±1.53	78.11±4.79	78.96±3.49	89.53±2.42	92.43±0.47
Co2L	**91.19±0.54**	**95.98±0.58**	81.25±4.94	88.44±2.48	90.74±3.07	90.61±0.44	59.58±4.12	68.4±3.94	97.75±0.38	94.71±0.84
DERPP	82.38±8.09	75.78±14.26	86.27±5.27	92.18±1.75	95.43±0.78	93.62±1.53	78.84±3.68	76.53±7.95	89.88±7.16	94.32±2.14
SimpleCIL	75.86±4.63	41.02±1.22	37.41±3.47	57.99±3.88	21.98±2.24	22.67±2.12	2.75±2.52	10.75±0.48	1.13±0.29	0.0±0.0
Ours	85.44±5.56	91.33±1.67	**93.88±0.77**	**96.26±0.24**	97.80±0.16	**96.41±0.97**	**92.65±0.59**	**90.90±0.98**	**98.93±0.32**	**98.30±0.36**

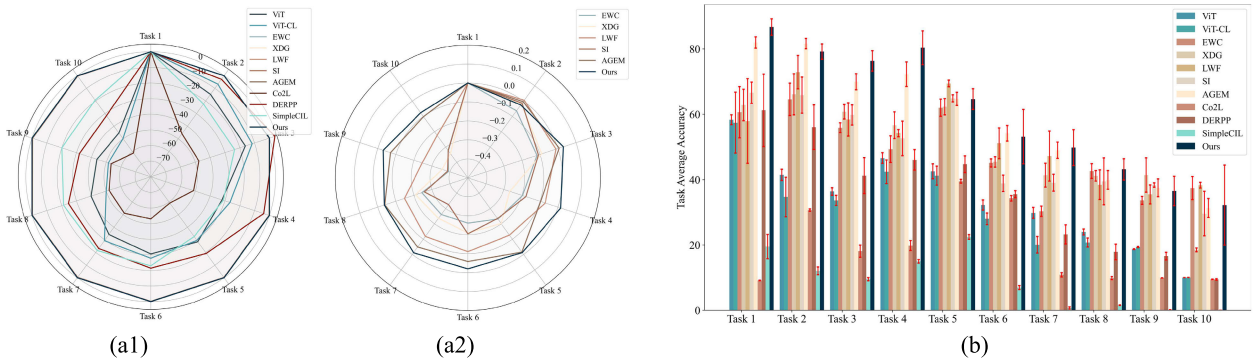


Fig. 16. BWT and TAA on the RSI dataset for comparison. (a-1) Ten-task BWT on the RSI dataset. (a-2) Ten-task BWT on the RSI dataset. (b) Task average accuracy comparison on the RSI dataset.

Similarly, compared to other methods, we perform better in handling the latest added data in accuracy and achieve more than 90% accuracy nine successive times. Specifically, in Task 3, our method improves accuracy by at least 3.14% compared to other methods, and in Task 7, our method improves accuracy by at least 6.55%.

As shown in Fig. 16(b), the average task accuracy plot for each stage of the model, we achieve a high average task accuracy across the board, which is weaker than the other compared methods only on Task 10, slightly lower than AGEM on Tasks 2 and 7, and lower than LWF on Task 5. The results indicate that our proposed model is still able to handle more complex datasets and maintain the ability to learn stably in the CL tasks.

As for the changes in BWT metrics among all methods on the RSI dataset, similar to the performance of ViT, ViT-CL, Co2L, and DERPP on the AID dataset, these methods get a weak TAA score in Fig. 16(b) and get the lowest BWT score in Fig. 16(a-1), which represents that these methods are easy to lose the learned latent knowledge. In Fig. 16(a-2), we can notice that most of our methods (black line) hold high BWT values this time and are only slightly lower than AGEM on Task 2, which indicates that our method can effectively tackle the historical data on the RSI dataset. From the confusion matrix in Fig. 17, it can be seen that all methods tend to predict the same class when confronted with future data, that is, they are unable to precisely discriminate future data, due to the larger size of the RSI data. While confronting historical data, our proposed method still maintains the best discriminative ability for historical data, which suggests that there is a superior advantage of our method in memorizing historical knowledge.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

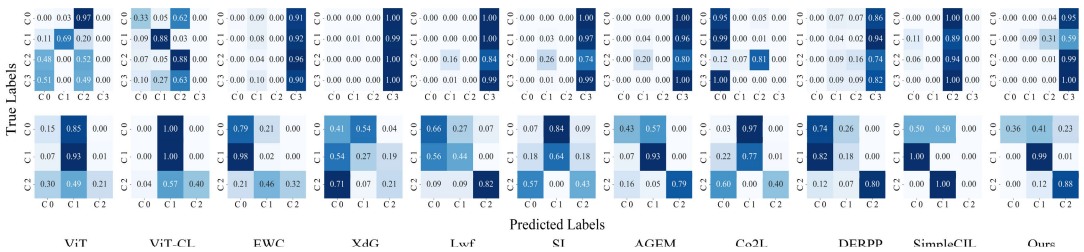


Fig. 17. Confusion matrix between our method and the comparison method on the RSI dataset.

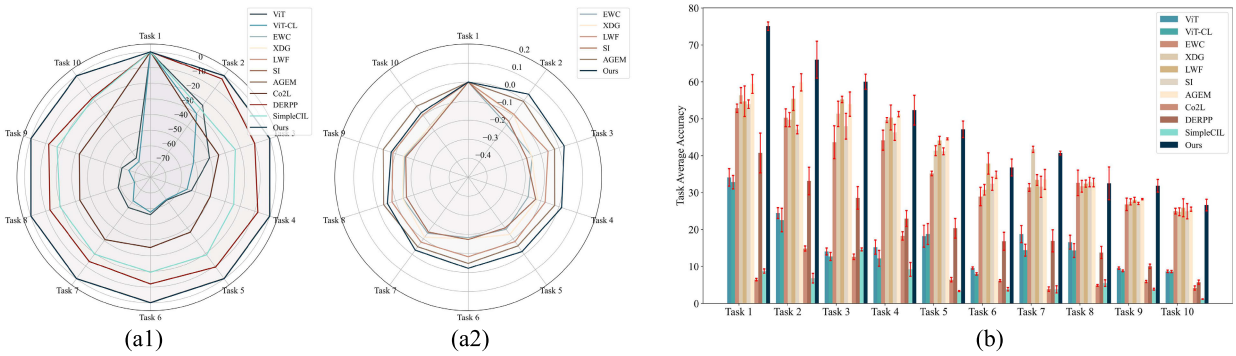


Fig. 18. BWT and TAA comparison on the NWPU dataset. (a-1) Ten-task BWT on the NWPU dataset. (a-2) Ten-task BWT on the NWPU dataset. (b) Task average accuracy comparison on the NWPU dataset.

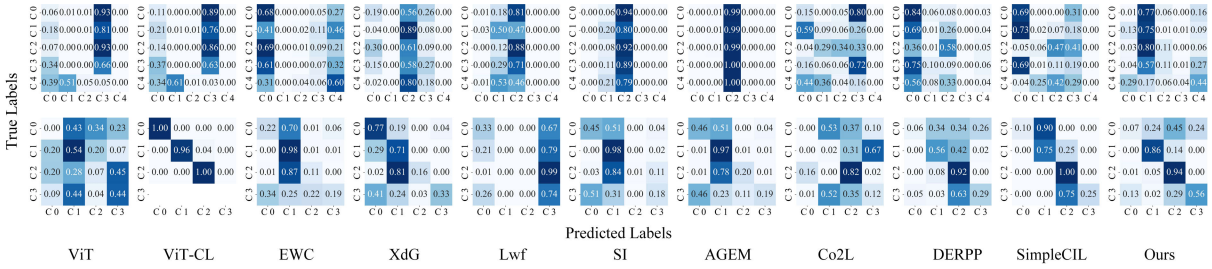


Fig. 19. Confusion matrix between our method and the comparison method on the NWPU dataset.

TABLE VI
COMPARISON RESULTS ONTHE NWPU DATASET

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Task9	Task10
ViT	93.04±0.73	96.19±0.34	90.36±1.02	85.6±1.73	84.88±1.35	71.05±1.14	88.19±2.87	83.43±1.23	74.76±3.29	86.52±3.27
ViT-cl	95.3±1.75	97.14±0.76	90.36±0.89	83.75±0.89	85.24±3.69	76.19±3.93	93.52±1.01	82.57±1.0	73.19±1.17	86.1±3.06
EWC	77.86±1.15	73.15±4.83	64.70±2.48	67.38±2.71	56.73±1.38	50.29±3.05	65.48±2.25	69.52±1.58	58.14±1.49	58.62±1.52
XdG	**78.51±1.97**	73.81±4.16	67.02±2.43	67.68±0.89	61.13±4.40	56.86±3.44	71.43±2.87	71.67±1.25	61.48±1.15	60.00±3.78
LwF	77.86±2.72	73.33±2.78	66.79±3.30	66.96±2.64	60.48±1.61	55.81±1.89	71.57±2.23	71.57±2.06	60.38±2.20	62.05±1.95
SI	77.62±2.43	73.03±3.45	66.61±2.84	68.45±0.55	59.17±3.42	54.10±3.04	70.38±2.06	71.19±1.78	59.86±2.44	60.43±4.58
AGEM	76.91±2.12	59.40±8.92	60.41±7.23	70.24±1.55	59.58±4.53	49.14±3.17	64.67±3.51	66.57±3.13	60.05±0.64	56.19±2.01
Co2L	64.4±3.13	73.87±0.66	50.48±7.93	44.23±5.77	15.42±1.24	24.9±2.04	32.38±5.05	48.71±2.22	37.0±5.42	41.76±5.58
DERPP	50.77±3.45	48.75±4.06	54.11±7.53	43.81±10.4	47.8±2.06	44.9±5.06	52.24±9.22	63.52±5.29	54.38±2.2	57.38±5.69
SimpleCIL	54.64±2.02	28.34±2.19	41.72±0.33	27.5±1.01	7.92±0.08	16.66±2.09	13.1±4.11	24.14±3.43	18.43±2.63	11.76±0.54
Ours	74.46±4.97	**75.77±5.79**	**71.85±1.47**	**70.83±5.3**	**63.63±5.29**	**62.33±3.75**	**77.38±4.88**	**75.38±1.77**	**62.90±3.00**	**67.91±1.02**

3) Comparison Experiment on the NWPU Dataset: In the NWPU dataset, the interclass similarity and intraclass variation problems affect the learning process significantly and provide a relatively low OA over the former two datasets. However, the results in Table VI exhibit that we have reached the best in nine successive times in ten-stage CL tasks on the NWPU dataset, and the accuracy exceeds 60% in all. It is only in the first stage that it is lower than the XDG method, with a 4.05% gap between our method and XDG, whereas, in Tasks 6, 7, and 10, our method improves at least 5.47%, 5.81%, and 5.86%, respectively. This proves that our method could deal with more complex datasets in a CL process. In Fig. 18(b), although the average accuracy of the model gradually decreases as the task continues, our method still maintains its advantage for achieving the best performance on all tasks in the sequence relative to other methods, which indicates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

P-VALUES OF OUR MODEL AND OTHER METHODS ONTHE AID DATASET (BOLD VALUES REPRESENT THAT THE PROPOSED METHOD IN THIS ARTICLE IS STATISTICALLY SIGNIFICANT WHEN P = 0.05)

	ViT	ViT-cl	EWC	XdG	LwF	SI	AGEM	Co2L	DERPP	SimpleCIL
OA	**1.19E-03**	**1.13E-03**	**5.67E-03**	**1.95E-02**	**8.34E-03**	**1.12E-02**	**8.20E-03**	**2.29E-04**	6.02E-01	**2.42E-05**
BWT	**3.74E-05**	**4.93E-05**	**1.70E-04**	**1.32E-03**	**1.41E-04**	**2.21E-04**	4.45E-01	**2.35E-04**	1.86E-01	**1.09E-03**
TAA	**3.32E-06**	**1.94E-06**	**1.26E-02**	1.42E-01	1.67E-01	**6.82E-03**	4.38E-01	**1.61E-07**	**1.25E-02**	**1.29E-07**

TABLE VIII

P-VALUES OF OUR MODEL AND OTHER METHODS ONTHE RSI DATASET (BOLD VALUES REPRESENT THAT THE PROPOSED METHOD IN THIS ARTICLE IS STATISTICALLY SIGNIFICANT WHEN P = 0.05)

	ViT	ViT-cl	EWC	XdG	LwF	SI	AGEM	Co2L	DERPP	SimpleCIL
OA	**6.24E-03**	**1.33E-02**	**1.55E-02**	5.37E-02	5.27E-02	7.04E-02	**5.12E-03**	5.21E-02	**1.20E-03**	**3.78E-05**
BWT	**1.05E-04**	**4.40E-04**	**1.94E-04**	**5.08E-04**	**2.48E-03**	**5.20E-04**	8.87E-02	**1.16E-05**	**1.60E-03**	**1.83E-04**
TAA	**3.24E-06**	**3.62E-06**	**1.16E-02**	**5.94E-03**	7.30E-02	**6.38E-03**	1.09E-01	**9.27E-05**	**2.70E-07**	**9.49E-07**

TABLE IX

P-VALUES OF OUR MODEL AND OTHER METHODS ONTHE NWPU DATASET (BOLD VALUES REPRESENT THAT THE PROPOSED METHOD IN THIS ARTICLE IS STATISTICALLY SIGNIFICANT WHEN P = 0.05)

	ViT	ViT-cl	EWC	XdG	LwF	SI	AGEM	Co2L	DERPP	SimpleCIL
OA	**4.32E-06**	**2.50E-06**	**2.31E-03**	**9.45E-03**	**2.66E-03**	**2.73E-03**	**2.78E-03**	**2.12E-04**	**1.32E-05**	**1.34E-06**
BWT	**3.77E-05**	**4.11E-05**	**6.15E-05**	**4.89E-05**	**1.09E-03**	**3.69E-05**	6.55E-01	**1.06E-05**	**1.11E-04**	**1.64E-04**
TAA	**9.79E-06**	**7.06E-06**	**1.54E-03**	**1.22E-02**	**3.08E-02**	**4.55E-03**	**1.29E-02**	**1.13E-05**	**1.68E-07**	**5.91E-06**

TABLE X

COMPARISON RESULTS OF THE MERGE-80 DATASET THAT MERGED THREE DATASETS (AID, RSI, AND NWPU)

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Task9	Task10
ViT	57.98±2.22	66.84±2.55	60.12±1.38	65.69±3.25	93.07±0.23	87.91±0.63	82.47±1.75	85.92±0.28	76.31±0.58	18.28±1.93
ViT-cl	56.96±3.15	66.5±3.1	59.59±3.18	68.67±5.53	94.27±0.69	88.0±0.42	86.23±1.33	87.95±1.63	76.85±2.5	22.03±1.08
EWC	35.42±0.97	54.16±2.66	50.52±3.15	62.50±4.18	89.84±1.69	79.95±5.35	82.03±2.78	56.77±2.95	43.49±3.28	51.04±3.27
XdG	35.94±3.88	54.69±2.92	49.22±2.92	60.42±1.47	92.45±0.74	83.07±0.74	89.06±1.10	62.24±4.48	45.05±1.33	55.21±6.46
LwF	36.72±3.37	53.12±2.92	49.48±2.05	57.55±3.51	91.67±0.37	84.12±1.61	88.54±0.37	62.24±2.24	47.40±2.24	53.91±5.85
SI	36.98±3.51	**57.03±4.47**	**52.60±3.68**	60.42±0.97	94.01±2.58	81.51±4.53	89.58±0.97	60.15±2.78	49.22±1.68	51.82±6.70
AGEM	**39.58±0.37**	54.42±3.01	46.10±5.56	**68.75±2.30**	81.51±2.58	78.65±4.25	79.69±1.91	52.60±3.27	38.80±3.27	59.64±4.10
Co2L	26.05±0.72	29.65±0.89	27.18±4.77	45.5±4.00	80.23±3.42	61.44±1.3	84.89±0.48	51.85±2.87	44.23±3.29	10.96±0.87
DERPP	23.56±4.07	42.74±4.18	35.91±3.32	30.87±6.72	81.32±4.15	**87.28±2.46**	80.18±1.19	**90.45±0.25**	**78.99±2.19**	22.13±1.8
SimpleCIL	34.67±2.04	31.35±0.42	9.72±1.56	35.4±1.99	57.89±0.5	20.52±3.13	34.93±5.23	22.0±1.39	6.55±0.48	4.17±1.22
Ours	27.9±3.11	53.40±4.93	40.28±13.06	66.32±2.15	**94.10±0.81**	86.19±0.74	**90.93±2.07**	63.21±2.54	51.04±2.47	**64.28±2.04**

that our model still provides superiority in complex CL sequences.

Similarly, in the BWT evaluation in Fig. 18(a-2), we can find that it lags behind AGEM on the last three stages of the task, yet our method consistently holds the highest BWT values on most of the tasks, which indicates that our method is able to cope with the historical data effectively. Following a resemblance to the results in the RSI dataset, all methods tend to predict the same class when confronted with future data, as shown in Fig. 19. When confronted with historical data, our proposed method still maintains the best ability to discriminate historical data, indicating that it is more advantageous to utilize our method in memorizing historical knowledge.

4) Statistical Analysis: We also conduct the statistical analysis on the OA, BWT, and TAA metrics to further demonstrate the robustness of the model results. Here, we chose to perform a paired t-test on the above three metrics based on the classification results at each stage in Tables IV–VI, and the paired t-test results on the AID, RSI, and NWPU datasets are shown in Tables VII–Tables IX. When the p-value is less than 0.05, we consider that there is a significant difference between the two algorithms, and it can be found that most of the p-values in these tables are less than 0.05; thus, the results prove that there is indeed a significant difference between our proposed method and the other comparative methods.

D. CL on Merged Datasets

1) CL on the Merge-80 Dataset: To validate the CL capacity of our model on complex, large-scale data streams, we merged the AID, RSI, and NWPU datasets. In this setup, unique categories were consolidated to form a total of 80 distinct categories for the CL task sequence. As shown in Table X, the results indicate that our method becomes increasingly capable of handling new data as the task stream updates. It is observed that our proposed model lagged slightly behind the SI and AGEM methods in the initial four-phase task (by 11.68%, 3.63%, 12.32%, and 2.43%, respectively). Similarly, in Tasks 6, 8, and 9, our method underperformed compared to DERPP, falling behind by 27.24% and 27.95% on Tasks 8 and 9, specifically. However, a distinct advantage emerges in the latter stages: while the accuracy of the AGEM method decreased

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

TABLE XI

CROSS-DATASET GENERALIZATION, WHERE A, R, AND N DENOTE THE AID, RSI, AND NWPU DATASETS, RESPECTIVELY. FA REPRESENTS FARMLAND, GFA IS GREEN FARMLAND, DE IS DESERT, BL IS BARELAND, MD IS MEADOW, BB IS BASEBALL FIELD, CH IS CHURCH, RS IS RAILWAY STATION, AP IS AIRPLANE, CR IS COMMERCIAL ARE A, AND "–" REPRESENTS THAT THE MODEL DOES NOT PERFORM THE ZERO-SHOT ABILITY

	FAa→GFAr	DEa→DEr	BLa→BLr	MDa→MDn	BBa→BBn	CHa→CHn	RSa→RSn	APr→APn	CRr→CRn
ViT	0.26±0.37	--	18.8±0.73	--	0.24±0.34	--	0.36±0.34	--	–
ViT-cl	–	--	–	--	--	--	–	--	–
EWC	15.0±7.07	53.9±38.74	60.42±10.62	15.87±19.18	8.33±11.79	--	24.07±6.93	31.88±26.64	11.98±11.58
XdG	26.98±25.89	76.1±14.96	**83.33±10.27**	15.00±17.80	20.51±7.25	--	20.51±14.50	35.56±13.7	17.71±13.75
LwF	11.11±9.78	51.57±27.14	30.00±29.44	16.67±12.47	12.82±18.13	6.25±5.1	41.03±26.34	40.0±27.22	27.6±7.34
SI	3.17±4.49	50.94±30.35	46.67±33.25	1.67±2.36	28.21±13.07	--	23.08±19.1	46.67±16.33	3.12±1.69
AGEM	28.79±4.29	86.81±6.44	78.43±7.34	**67.68±17.9**	27.27±7.42	9.26±5.24	20.0±13.06	**50.0±13.61**	9.38±2.55
Co2l	–	--	–	--	1.43±0.0	--	–	--	–
DERPP	–	--	–	--	--	--	–	8.81±12.46	0.24±0.34
SimpleCIL	9.11±11.8	22.62±10.72	17.44±23.85	32.14±4.21	0.24±0.33	1.19±1.68	4.05±4.3	2.38±1.22	0.0±0.0
Ours	**50.78±7.25**	**97.11±0.70**	70.74±11.65	48.57±4.77	**42.86±13.71**	**33.57±7.89**	**50.24±5.03**	11.19±6.4	**43.1±3.32**

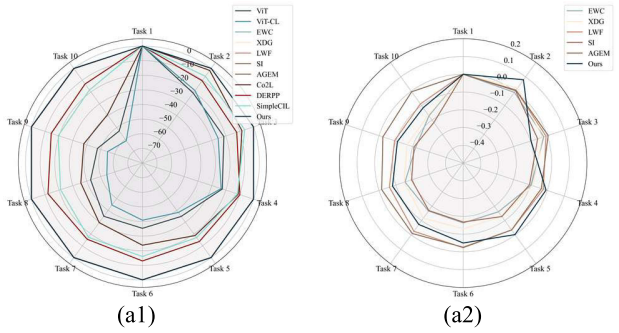


Fig. 20. BWT results on the Merge-80 dataset. (a-1) Ten-task BWT on the Merge-80 dataset. (a-2) Ten-task BWT on the Merge-80 dataset.

significantly and DERPP declined by 56.86% in Task 10, our method maintained stable performance throughout the last six consecutive tasks. Notably, in the fifth, seventh, eighth, and ninth tasks, our method improved by at least 10% compared to AGEM, demonstrating superior long-term stability.

We further utilized BWT to measure the memorization ability of the model. As illustrated in Fig. 20, although the AGEM method exhibits the highest BWT values from Task 6 to Task 10, this is attributed to its significant decline in overall performance at these stages rather than effective retention. In contrast, our model achieves high BWT values across most stages alongside increasing accuracy, indicating that it minimizes the loss of previously learned knowledge. In addition, we utilized Grad-CAM to visualize the model's attention during this process. As shown in Fig. 21(b), comparative methods tend to focus merely on regional information, such as specific parts of a City Building or Port. Conversely, our method effectively captures global, detailed semantic information, allowing it to accurately determine corresponding categories even within a long-sequence task.

2) Cross-Dataset Generalization Inference: To further verify the ability of our proposed model to handle cross-domain shifts, we conducted a zero-shot generalization validation using the duplicate categories that were excluded from the training process. In this setup, the model is fully trained on a source dataset (e.g., Farmland in AID) and subsequently evaluated on a distinct target dataset (e.g., Green Farmland in RSI) without any additional fine-tuning. This is a challenging scenario; as shown in Fig. 22, the same scene categories across datasets differ significantly in color, scale, and internal structure. For instance, the AID and NWPU datasets differ in the Railway Station background information, while airplanes in RSI and NWPU exhibit inconsistent scales.

The quantitative results of this generalization test are presented in Table XI. It is notable that ViT, ViT-CL, Co2L, and DERPP perform weakest on cross-dataset generalization, with accuracy on various categories approaching 0%. This indicates that these methods are unable to leverage current knowledge to reason about unseen domains. In contrast, our method achieves the highest accuracy on most categories; for example, it achieves 97.11% accuracy on the desert category in the RSI dataset without using any annotated samples from that target domain. Similarly, for the church category, our method maintains reasonable discrimination capabilities where comparative methods perform poorly.

Qualitative attention analysis further supports these findings. As shown in Fig. 21(a) for unseen Island and Overpass categories and in Fig. 23 for cross-dataset recognition, our model accurately identifies critical semantic regions—such as the center island area or the specific structure of an overpass. While other approaches tend to focus on background noise (e.g., the area around a river) or localized edges, our method is effective in obtaining sufficient global information. This is particularly evident in dense residential areas with complex internal structures, where our method prioritizes representative areas over simple localization.

E. Large-Scale CL on the Merge-90 Dataset and Cross-Seasonal Continual Learning

1) Cross-Season CL: Considering the variation of environmental factors such as seasons and illumination conditions during the updating process of remote sensing data streams, we performed a CL task based on the SEASONET dataset to validate the robustness of our approach on these highly varied data.

Based on the collected SEASONET dataset comprising cross-seasonal scenarios, we designed a specific task sequence to evaluate performance under seasonal and illumination changes. As shown in Table XII, we demonstrate the accuracy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

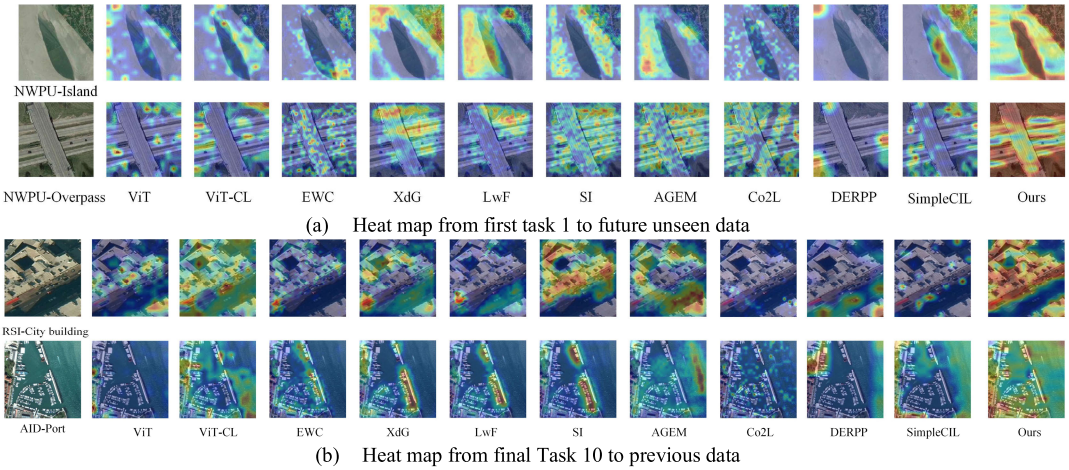


Fig. 21. Heat map comparison on different steps. (a) is used to demonstrate the performance on future tasks, and (b) is utilized to display the memorization of learned knowledge.

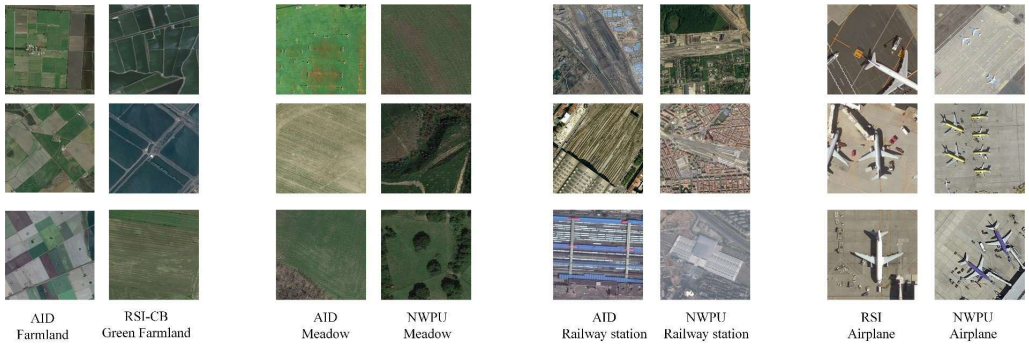


Fig. 22. Comparison of similar categories between different datasets.

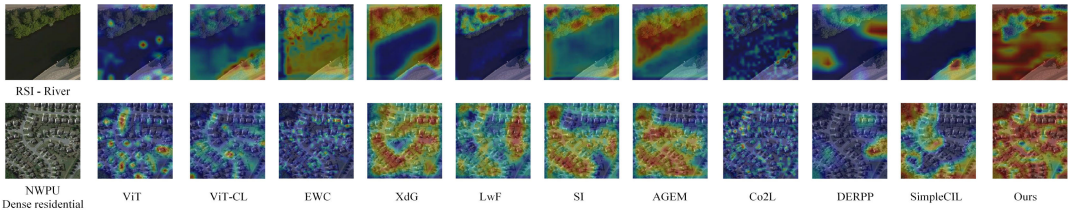


Fig. 23. Heat map demonstrate of the final step task on unseen images.

TABLE XII
CROSS-SEASON CL TASKS: THE BOLDED AND ITALICIZED SEASON INDICATES THE CURRENT TASK

	Task1	Task2		Task3			Task4			
	Spring	Spring	***Summer***	Spring	Summer	***Fall***	Spring	Summer	Fall	***Winter***
ViT	76.72±1.22	0.0±0.0	77.0±0.54	0.0±0.0	0.06±0.08	79.89±1.1	0.0±0.0	0.06±0.08	1.17±0.82	81.44±1.32
ViT-cl	83.06±1.73	0.89±0.63	77.5±1.71	0.11±0.16	0.0±0.0	81.28±0.91	0.0±0.0	0.0±0.0	0.61±0.48	84.11±1.22
EWC	34.11±8.1	25.26±11.49	36.46±9.3	27.08±7.37	33.85±11.24	39.06±8.86	28.12±10.9	30.73±10.53	33.33±9.3	38.54±11.35
XdG	39.58±5.31	12.5±2.92	39.32±8.49	9.38±1.28	21.61±4.53	32.81±16.35	9.38±2.92	17.45±9.81	18.49±11.4	25.78±19.05
LwF	40.1±3.9	41.15±3.9	51.82±1.61	35.68±1.61	48.96±5.31	51.04±5.2	32.55±2.42	32.29±6.52	41.41±6.28	46.88±8.58
SI	35.94±12.12	28.91±3.55	33.33±7.03	29.69±7.1	33.59±8.93	35.94±4.98	23.7±5.2	29.43±12.89	33.85±12.9	35.68±4.53
AGEM	42.45±3.9	41.15±7.48	46.35±10.53	45.31±4.47	**53.39±3.21**	51.56±6.15	47.66±4.47	**51.3±3.21**	49.48±5.79	49.22±2.92
Co2L	**60.5±1.11**	12.67±1.65	54.89±2.07	5.39±0.7	8.17±0.0	61.56±0.61	7.56±0.98	3.22±0.21	4.28±0.31	62.72±1.0
DERPP	44.94±3.1	34.5±5.74	52.06±6.84	30.17±7.17	36.61±11.1	47.5±14.39	32.72±3.0	34.5±1.57	42.28±3.0	58.94±5.69
SimpleCIL	24.83±0.76	3.61±0.42	22.78±1.74	2.33±1.08	5.55±2.65	23.11±2.99	2.33±0.47	4.22±3.26	9.83±4.26	13.44±0.92
Ours	56.11±0.28	**51.89±4.82**	**61.17±2.48**	**50.0±2.13**	52.72±2.5	**69.89±0.67**	**48.33±1.08**	49.06±4.97	**61.44±4.57**	**69.44±1.26**

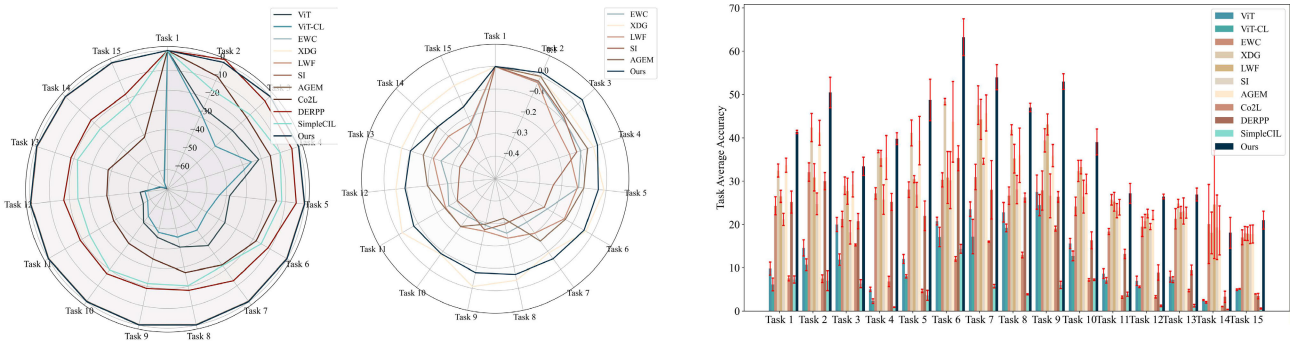
of comparative methods alongside our proposed method on the newest season during the CL process, as well as the retention accuracy on previously experienced seasons. Results indicate that, with the exception of slightly lower accuracy in the first

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

TABLE XIII

COMPARISON RESULTS OF LARGE-SCALE CL TASKS ON THE MERGE-90 DATASET THAT COMBINED FOUR DIFFERENT DATASETS

	Task1	Task2	Task3	Task4	Task5	Task6	Task7	Task8	Task9	Task10	Task11	Task12	Task13	Task14	Task15
ViT	56.62±7.95	77.39±2.5	80.2±5.21	32.48±6.64	73.23±8.03	95.15±0.37	91.15±1.85	87.92±1.71	90.0±0.77	88.3±0.55	87.82±0.73	78.25±3.86	86.39±2.25	22.94±4.22	74.26±2.14
ViT-cl	58.46±7.05	79.4±4.97	71.15±10.33	26.1±2.12	72.8±2.57	91.89±4.84	87.08±3.45	85.26±3.39	90.08±2.19	89.52±0.51	87.26±3.41	79.44±1.13	84.33±3.42	25.7±2.36	76.71±1.39
EWC	30.73±5.79	54.95±2.24	34.9±4.79	54.17±9.47	52.6±8.1	76.04±1.95	87.24±2.58	64.58±0.97	69.79±8.03	70.05±7.0	37.24±4.83	36.98±0.97	43.75±2.92	19.79±6.13	41.67±5.35
XdG	33.33±3.63	58.33±1.47	37.5±8.93	44.53±6.75	52.86±6.29	75.26±1.61	82.03±5.06	73.96±2.66	79.17±1.84	60.42±3.85	44.27±6.13	34.9±2.05	59.64±3.85	19.53±3.83	37.5±2.78
LwF	43.49±2.24	53.39±8.2	45.05±3.63	51.04±4.15	55.21±5.43	87.24±3.85	85.42±6.29	72.92±4.87	81.77±1.95	72.92±3.51	51.3±5.75	45.57±1.47	57.81±3.98	19.53±2.3	43.23±0.97
SI	39.06±3.98	55.21±8.57	32.03±8.93	54.17±3.74	59.11±2.58	86.2±2.24	75.78±18.79	61.72±2.78	76.30±2.88	75.26±4.48	54.43±1.33	44.53±2.3	60.16±3.19	16.93±2.88	44.01±2.42
AGEM	36.46±0.37	57.55±6.7	42.97±5.52	55.73±3.63	56.25±3.98	82.03±1.1	73.18±8.42	70.31±13.5	72.4±10.6	57.29±2.24	41.41±4.82	33.33±6.03	52.34±5.06	17.97±3.55	39.06±6.63
Co2L	36.85±2.75	38.36±0.31	29.2±8.93	12.76±3.54	32.56±6.04	86.55±1.79	86.36±3.3	68.17±4.86	83.01±2.21	63.29±1.73	42.82±3.9	39.01±2.98	55.28±3.57	15.25±0.48	**59.38±0.4**
DERPP	25.8±0.24	44.39±6.27	32.91±2.36	42.67±4.93	54.06±5.22	81.1±2.91	66.57±16.67	79.09±3.42	71.24±11.57	66.98±5.46	50.32±6.69	44.29±3.21	62.54±0.9	23.79±5.86	50.83±9.17
SimpleCIL	41.12±0.92	32.33±3.84	26.85±2.38	8.1±1.4	20.38±2.47	56.36±1.93	54.81±4.66	19.56±1.64	30.72±11.77	31.22±1.28	25.75±4.13	8.61±0.3	13.53±4.06	4.04±0.72	8.94±1.92
Ours	**44.64±2.87**	**61.31±3.8**	**45.51±3.1**	**59.43±4.79**	**72.45±4.49**	**94.32±1.0**	**93.37±0.38**	**87.35±1.52**	**91.04±1.59**	**81.47±3.13**	**60.83±0.85**	**60.36±3.4**	**70.67±0.66**	**26.66±4.87**	58.08±1.22



(a-1) 10-task BWT on Merge-90 dataset (a-2) 10-task BWT on Merge-90 dataset (b) Task average accuracy comparison on Merge-90 dataset

Fig. 24. BWT and TAA on the Merge-90 dataset for CL comparison. (a-1) Ten-task BWT on the Merge-90 dataset. (a-2) Ten-task BWT on the Merge-90 dataset. (b) Task average accuracy comparison on the Merge-90 dataset.

task, our method not only achieves optimal performance on the current season but also maintains the highest accuracy in recognizing scenes from historical seasons.

In particular, for standard ViT and ViT-CL methods, the accuracy performance declined extremely rapidly regarding previous seasons. This indicates that these methods tend to bias heavily toward exploiting the latest knowledge, neglecting the stability required to maintain remote sensing concepts from previous seasons. Compared to other CL methods, our model performs significantly better on cross-season tasks. Specifically, our model is inferior to Co2L (by 4.39%) only in the first task. However, in all subsequent tasks, our method obtains the optimal accuracy on the newest season step and performs robustly on already experienced data. For instance, in Task 4, our model improves accuracy on "Spring" scenes by at least 0.67% and on "Fall" scenes by at least 11.96% over other methods. This demonstrates that our proposed model can stably maintain the memory of historical knowledge even under significant cross-season domain shifts.

2) Large-Scale 15-Stage Task Sequence: To further adapt the model to longer CL task sequences and evaluate its limits, we integrated the SEASONET dataset with the previous three datasets (AID, RSI, and NWPU). We designed a challenging 15-stage CL task with six scene types in each stage, which serves to validate the model's ability to learn continuously across varying resolutions and diverse data sources.

In this large-scale CL scenario, as shown in Table XIII, when comparing against standard scene classification models, our method gradually approaches the upper-bound accuracy on the latest tasks and even exceeds standard performance on Tasks 4, 7, 9, and 14. When compared with other CL methods, our model achieves the highest accuracy performance on the vast majority of tasks. Among the baselines, LwF, SI, and DERPP are able to maintain relatively stable performance during the learning process, whereas other methods are prone to severe degradation. For instance, the performance of EWC directly decreases by 32.81% at Task 10, while Co2L improved accuracy by 53.99% at Task 5. Notably, in Task 14, since the data source shifts from high-resolution imagery to Sentinel-2 scenes, the accuracy of all models suffered a significant decline. However, as the models adapted to the data, only our method, Co2L, and DERPP accelerated the recovery of accuracy. This indicates that our method is able to effectively acquire representative information of the current task with the assistance of the visual description memory mechanism.

Finally, we utilize BWT and average task accuracy to measure the memory capability of the models regarding historical knowledge. As shown in Fig. 24, although regular scene classification methods can achieve high performance quickly in the current task, they lack the ability to memorize historical knowledge in longer sequence tasks. While CL baselines are more capable in this regard, our proposed model is significantly more stable. As shown in Fig. 24(a-2), although XdG exhibits higher BWT values than our method on multiple subsequent tasks, this is an artifact of the XdG method achieving much lower overall performance during learning [as shown in Table XIII and Fig. 24(b)]. Consequently, the absolute accuracy and average task accuracy of XdG are significantly lower than those of our method. This indicates a significant advantage of our method in balancing the learning of new concepts and the memorization of historical knowledge, even in large-scale CL tasks involving resolution, season, and illumination variations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

VI. Discussion

The proposed DVDM framework represents a significant paradigm shift compared to existing remote sensing classification methods, particularly in addressing the stability–plasticity dilemma. Traditional deep learning models, such as standard ViT, typically map images to a fixed latent space. This static nature inevitably leads to catastrophic forgetting when the underlying data distribution shifts due to the emergence of new classes or domains. In sharp contrast, DVDM constructs a dynamic visual vocabulary that evolves alongside the data stream. This mechanism allows the model to interpret novel scenes by recombining existing visual primitives and adaptively creating new ones, thereby offering a significantly more plastic and scalable representation. Furthermore, while traditional BoVWs methods utilize discrete features to describe scenes, they often lack the semantic depth of deep representations and remain inherently static. DVDM effectively bridges this gap by integrating the explicit interpretability of visual words with the high-level representational power of Transformer-based features. This synergy ensures robust performance and generalization capabilities, even under complex scenarios involving simultaneous class- and domain-incremental challenges.

Despite these advancements, the current DVDM framework has limitations. Continuous K-Means updates introduce computational overhead that increases with task sequences and vocabulary size K, while performance in long-sequence scenarios remains sensitive to memory buffer capacity.

VII. Conclusion

In this article, we proposed the DVDM framework to effectively address the dual challenges of class incremental and domain shift learning in remote sensing data streams. By dynamically clustering deep feature patches into a universal visual vocabulary, DVDM constructs robust semantic descriptions that bridge the gap between evolving tasks. Furthermore, we integrated a gradient-constrained memory mechanism to mathematically bound the optimization trajectory, thereby successfully navigating the stability–plasticity dilemma—minimizing the loss of historical knowledge while maintaining the flexibility to learn from new data.

Extensive experiments were conducted on a comprehensive set of benchmarks, including standard datasets (AID, RSI, and NWPU), a merged large-scale stream, and a challenging cross-season sequence constructed from SEASONET. The results demonstrate that our method significantly outperforms existing baselines in handling distribution shifts caused by seasonal and sensor variations. DVDM effectively enhances scene comprehension and promotes the retention of historical knowledge even under complex environmental conditions, proving its robustness in balancing plasticity and stability across long-sequence tasks.

In future work, addressing the high cost of acquiring annotated samples remains a priority. We aim to extend DVDM toward label-efficient paradigms, specifically exploring few-shot or even zero-shot CL to reduce dependence on large-scale annotations. In addition, we plan to refine the semantic linkage capability of our visual vocabulary to better generalize to unseen categories. Finally, we intend to adapt the proposed CL strategy to other dense prediction tasks, such as change detection and land-cover classification, to further tackle the challenges of ever-increasing remote sensing data with reduced training costs.

References

[1] J. Wang, Y. Zhong, A. Ma, Z. Zheng, Y. Wan, and L. Zhang, "LoveNAS: Towards multi-scene land-cover mapping via hierarchical searching adaptive network," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 265–278, Mar. 2024, doi: 10.1016/j.isprsjprs.2024.01.011.

[2] J. Ge, H. Tang, N. Yang, and Y. Hu, "Rapid identification of damaged buildings using incremental learning with transferred data from historical natural disaster cases," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 105–128, Jan. 2023, doi: 10.1016/j.isprsjprs.2022.11.010.

[3] Y. Chen, Q. Weng, L. Tang, L. Wang, H. Xing, and Q. Liu, "Developing an intelligent cloud attention network to support global urban green spaces mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 197–209, Apr. 2023, doi: 10.1016/j.isprsjprs.2023.03.005.

[4] K. Nogueira, O. A. B. Penatti, and J. A. dos Santos, "Towards better exploiting convolutional neural networks for remote sensing scene classification," *Pattern Recognit.*, vol. 61, pp. 539–556, Jan. 2017, doi: 10.1016/j.patcog.2016.07.001.

[5] L. Fang, Y. Yan, J. Yue, and Y. Deng, "Toward the vectorization of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5518214, doi: 10.1109/TGRS.2023.3299154.

[6] W. Zhou, Z. Shao, and Q. Cheng, "Deep feature representations for high-resolution remote sensing scene classification," in *Proc. 4th Int. Workshop Earth Observ. Remote Sens. Appl. (EORSA)*, Jul. 2016, pp. 338–342.

[7] G. Cheng, C. Ma, P. Zhou, X. Yao, and J. Han, "Scene classification of high resolution remote sensing images using convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 767–770.

[8] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.

[9] W. Huang, Y. Shi, Z. Xiong, Q. Wang, and X. X. Zhu, "Semi-supervised bidirectional alignment for remote sensing cross-domain scene classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 192–203, Jan. 2023, doi: 10.1016/j.isprsjprs.2022.11.013.

[10] X. Lu, H. Sun, and X. Zheng, "A feature aggregation convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7894–7906, Oct. 2019.

[11] Z. Zheng, Y. Zhong, Y. Su, and A. Ma, "Domain adaptation via a task-specific classifier framework for remote sensing cross-scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620513, doi: 10.1109/TGRS.2022.3151689.

[12] S. Dohare, J. F. Hernandez-Garcia, Q. Lan, P. Rahman, A. R. Mahmood, and R. S. Sutton, "Loss of plasticity in deep continual learning," *Nature*, vol. 632, no. 8026, pp. 768–774, Aug. 2024, doi: 10.1038/s41586-024-07711-7.

[13] V. Araujo, J. Hurtado, A. Soto, and M.-F. Moens, "Entropy-based stability-plasticity for lifelong learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 3720–3727.

[14] D. Kim and B. Han, "On the stability-plasticity dilemma of class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20196–20204.

[15] T. Zhang, B. Ghanem, S. Liu, C. Xu, and N. Ahuja, "Low-rank sparse coding for image classification," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 281–288.

[16] E. Li, P. Du, A. Samat, Y. Meng, and M. Che, "Mid-level feature representation via sparse autoencoder for remotely sensed scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 3, pp. 1068–1081, Mar. 2017.

[17] Z. Fang, W. Li, J. Zou, and Q. Du, "Using CNN-based high-level features for remote sensing scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 2610–2613.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.

[18] Q. Xu et al., "Hierarchical spiking-based model for efficient image classification with enhanced feature extraction and encoding," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9277–9285, Jul. 2024.
[19] H. Li et al., "Task-specific fine-tuning via variational information bottleneck for weakly-supervised pathology whole slide image classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7454–7463.
[20] A. Ganesh et al., "Why is public pretraining necessary for private model training?," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 10611–10627.
[21] G. Cheng, Z. Li, X. Yao, L. Guo, and Z. Wei, "Remote sensing image scene classification using bag of convolutional features," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1735–1739, Oct. 2017, doi: 10.1109/LGRS.2017.2731997.
[22] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.
[23] U. Muhammad, W. Wang, S. P. Chattha, and S. Ali, "Pre-trained VGGNet architecture for remote-sensing image scene classification," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 1622–1627.
[24] W. Zhang, P. Tang, and L. Zhao, "Remote sensing image scene classification using CNN-CapsNet," *Remote Sens.*, vol. 11, no. 5, p. 494, Feb. 2019, doi: 10.3390/rs11050494.
[25] Z. Mai, R. Li, J. Jeong, D. Quispe, H. Kim, and S. Sanner, "Online continual learning in image classification: An empirical survey," *Neurocomputing*, vol. 469, pp. 28–51, Jan. 2022, doi: 10.1016/j.neucom.2021.10.021.
[26] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, 2017.
[27] R. Aljundi, F. Babiloni, M. Elhoseiny, M. Rohrbach, and T. Tuytelaars, "Memory aware synapses: Learning what (not) to forget," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 144–161.
[28] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. 34th Int. Conf. Mach. Learn.*, 2017, pp. 3987–3995.
[29] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2018.
[30] S. Jung, H. Ahn, S. Cha, and T. Moon, "Continual learning with node-importance based adaptive group sparse regularization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 3647–3658.
[31] J. Serrá, D. Surís, M. Miron, and A. Karatzoglou, "Overcoming catastrophic forgetting with hard attention to the task," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 4548–4557.
[32] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7765–7773.
[33] O. Ostapenko, P. Rodríguez, M. Caccia, and L. Charlin, "Continual learning via local module composition," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 30298–30312.
[34] L. Wang et al., "Incorporating neuro-inspired adaptability for continual learning in artificial intelligence," *Nature Mach. Intell.*, vol. 5, no. 12, pp. 1356–1368, Nov. 2023, doi: 10.1038/s42256-023-00747-w.
[35] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "ICaRL: Incremental classifier and representation learning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5533–5542.
[36] D. López-Paz and M. Ranzato, "Gradient episodic memory for continual learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 6467–6476.
[37] C. Wu, L. Herranz, X. Liu, Y. Wang, J. v. d. Weijer, and B. Raducanu, "Memory replay GANs: Learning to generate images from new categories without forgetting," *Adv. Neural Inf. Process. Syst.*, pp. 5962–5972, 2018.
[38] J. Bang, H. Kim, Y. Yoo, J. Ha, and J. Choi, "Rainbow memory: Continual learning with a memory of diverse samples," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New York, NY, USA, Jun. 2021, pp. 8214–8223, doi: 10.1109/CVPR46437.2021.00812.
[39] G. Petit, A. Popescu, H. Schindler, D. Picard, and B. Delezoide, "FeTrIL: Feature translation for exemplar-free class-incremental learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 3900–3909, doi: 10.1109/WACV56688.2023.00390.
[40] N. Ammour, Y. Bazi, H. Alhichri, and N. Alajlan, "Continual learning approach for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3019071.
[41] R. Peng, W. Zhao, K. Li, F. Ji, and C. Rong, "Continual contrastive learning for cross-dataset scene classification," *Remote Sens.*, vol. 14, no. 20, p. 5105, Oct. 2022, doi: 10.3390/rs14205105.
[42] M. A. Ma'Sum, M. Pratama, R. Savitha, L. Liu, Habibullah, and R. Kowalczyk, "Unsupervised few-shot continual learning for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4707214, doi: 10.1109/TGRS.2024.3445887.
[43] L. Zhao et al., "Continual learning for remote sensing image scene classification with prompt learning," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023, doi: 10.1109/LGRS.2023.3328981.
[44] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017, doi: 10.1109/TGRS.2017.2685945.
[45] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017, doi: 10.1109/JPROC.2017.2675998.
[46] H. Li et al., "RSI-CB: A large-scale remote sensing image classification benchmark using crowdsourced data," *Sensors*, vol. 20, no. 6, p. 1594, Mar. 2020.
[47] D. Koßmann, V. Brack, and T. Wilhelm, "Seasonet: A seasonal scene classification, segmentation and retrieval dataset for satellite imagery over Germany," in *Proc. IGARSS - IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 243–246, doi: 10.1109/IGARSS46834.2022.9884079.
[48] M. Bi, M. Wang, Z. Li, and D. Hong, "Vision transformer with contrastive learning for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 738–749, 2023, doi: 10.1109/JSTARS.2022.3230835.
[49] N. Y. Masse, G. D. Grant, and D. J. Freedman, "Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization," *Proc. Nat. Acad. Sci. USA*, vol. 115, no. 44, pp. E10467–E10475, Oct. 2018, doi: 10.1073/pnas.1803839115.
[50] A. Chaudhry, M. Ranzato, M. Rohrbach, and M. Elhoseiny, "Efficient lifelong learning with A-GEM," in *Proc. 7th Int. Conf. Learn. Represent.*, 2018, pp. 1–20.
[51] P. Buzzega, M. Boschini, A. Porrello, D. Abati, and S. Calderara, "Dark experience for general continual learning: A strong, simple baseline," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 15920–15930.
[52] H. Cha, J. Lee, and J. Shin, "Co2L: Contrastive continual learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9496–9505, doi: 10.1109/ICCV48922.2021.00938.
[53] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, and Z. Liu, "Revisiting class-incremental learning with pre-trained models: Generalizability and adaptivity are all you need," *Int. J. Comput. Vis.*, vol. 133, no. 3, pp. 1012–1032, Mar. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 14:47:28 UTC from IEEE Xplore. Restrictions apply.